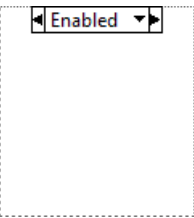
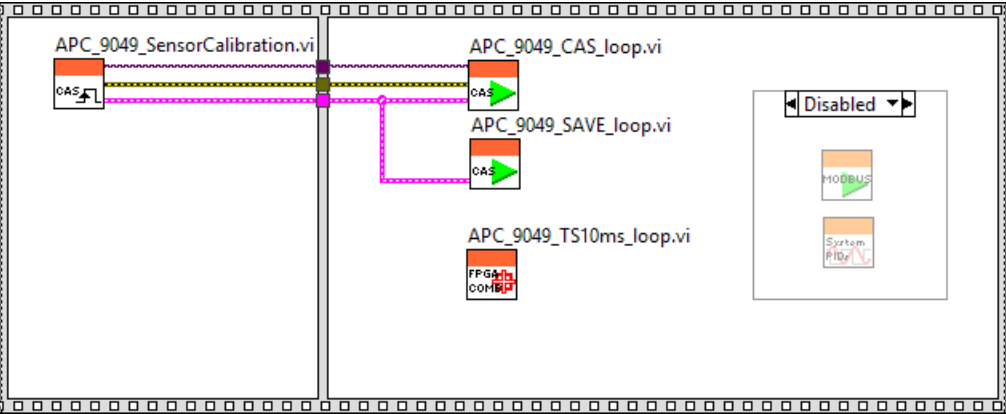
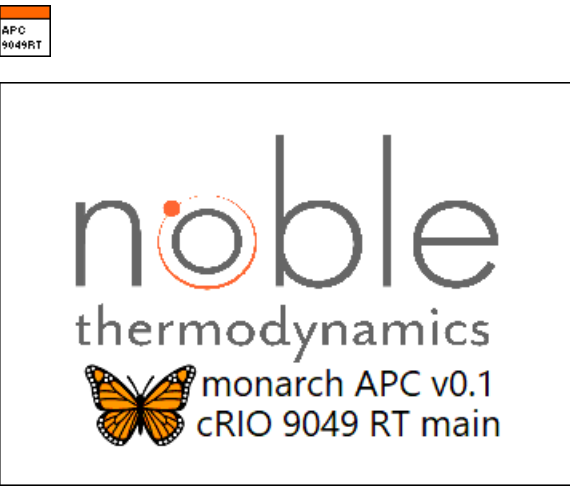


APC_9049_RT_main.vi

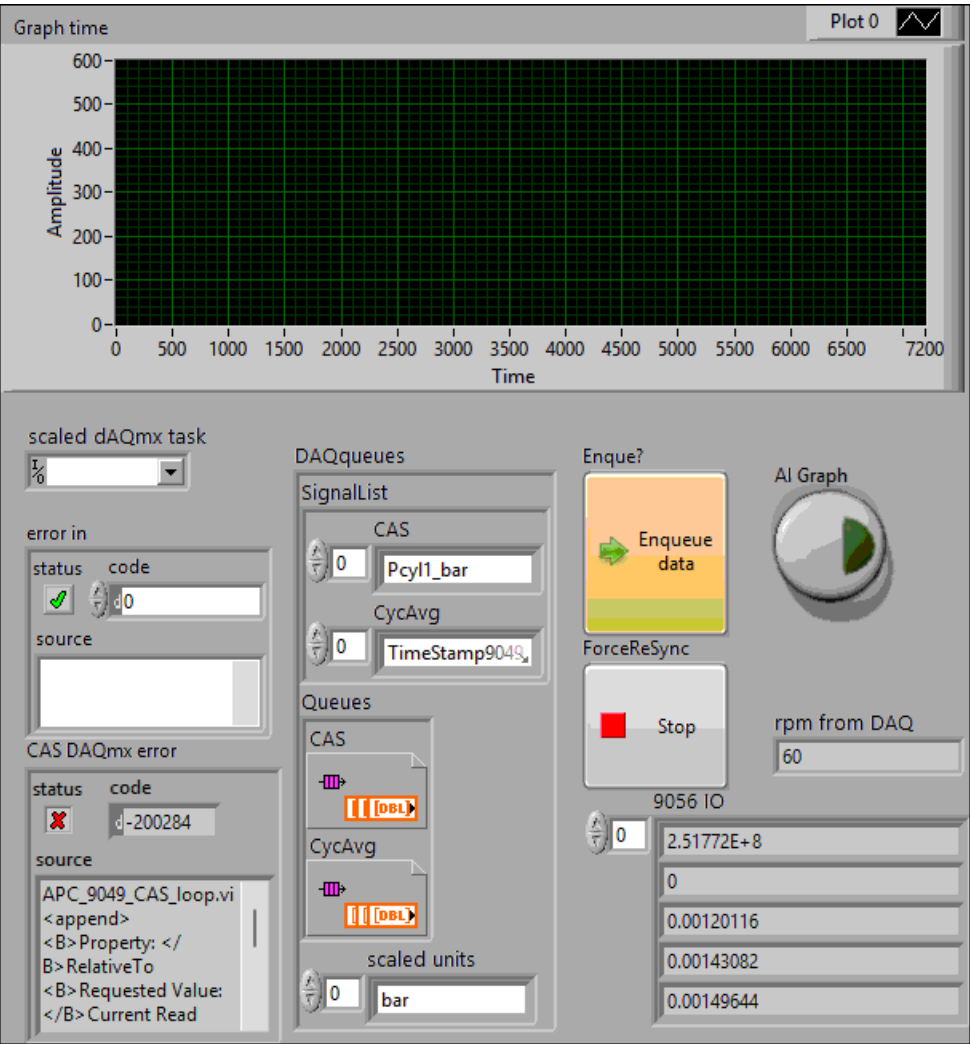
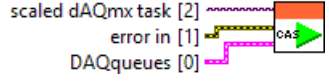


- APC_9049_CAS_loop.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_CAS_loop.vi
- APC_9049_SAVE_loop.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_SAVE_loop.vi
- APC_9049_TS10ms_loop.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_TS10ms_loop.vi
- APC_9049_SensorCalibration.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_SensorCalibration.vi
- APC_9049_MODBUSTCP_HealingLoop.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_MODBUSTCP_HealingLoop.vi
- APC_9049_TS100ms_loop.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_TS100ms_loop.vi

APC_9049_CAS_loop.vi

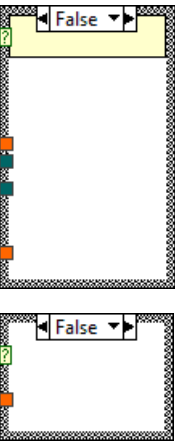
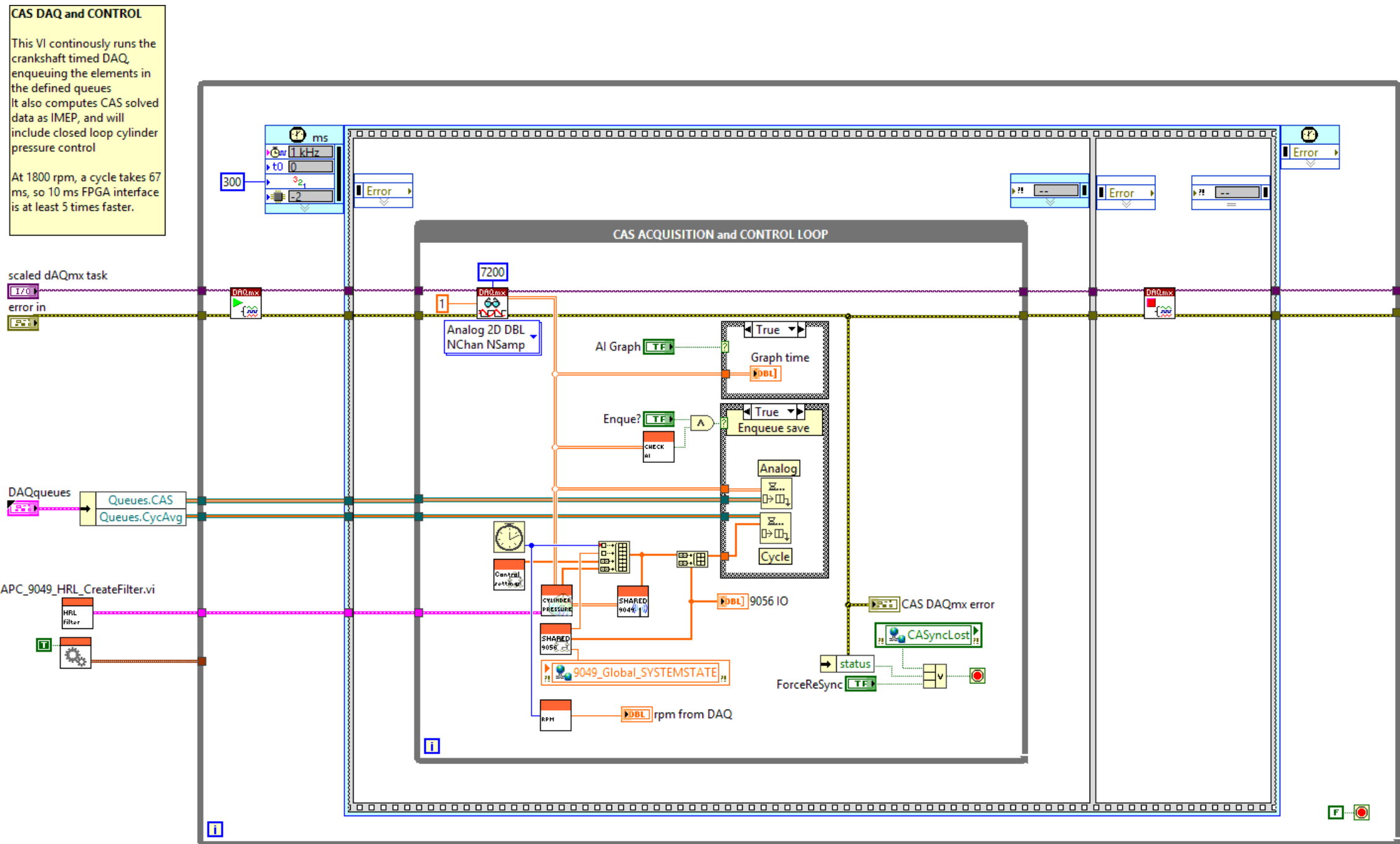
This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

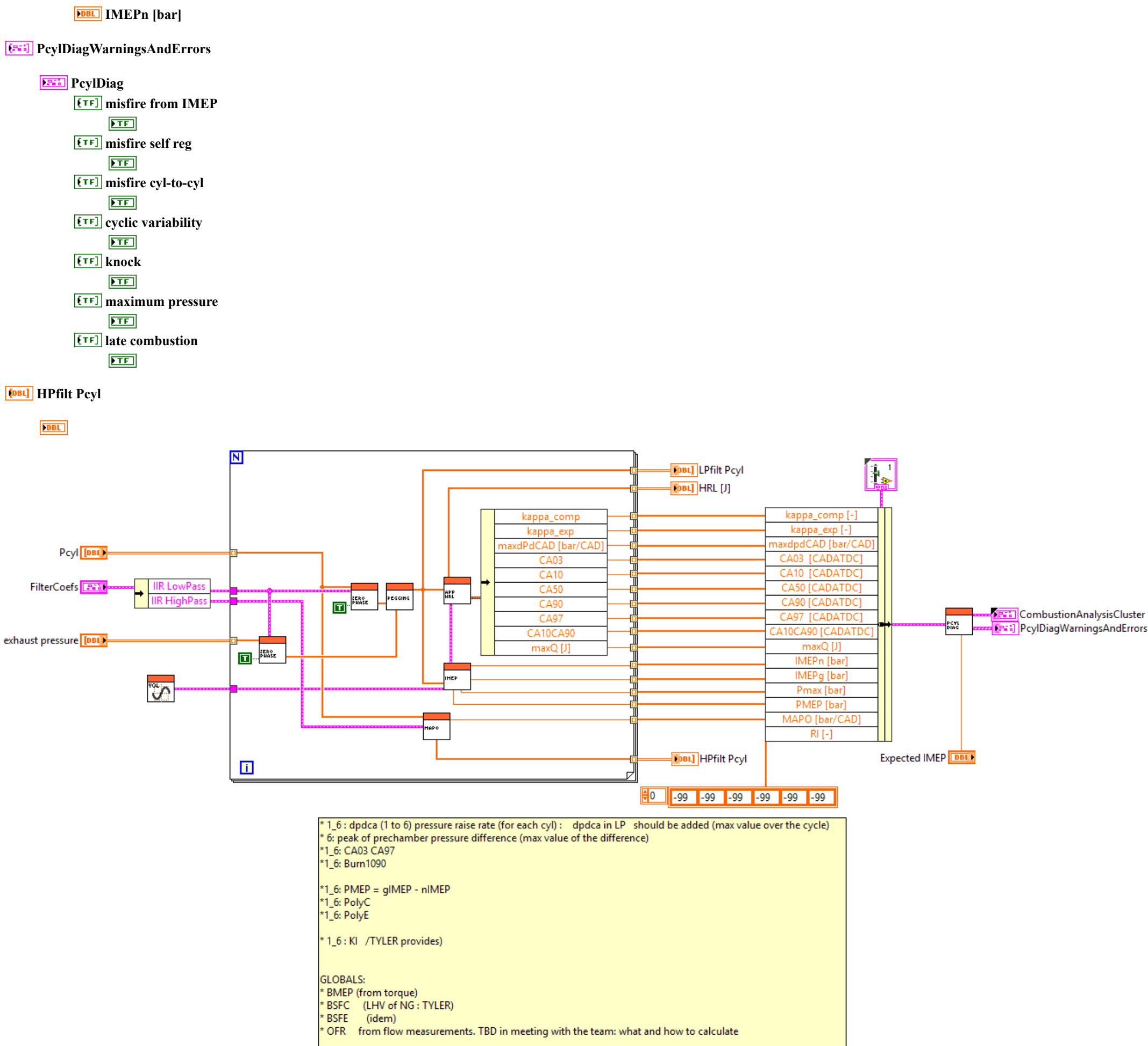
The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.



- scaled dAQmx task
 - error in
 - status
 - code
 - source
- error in can accept error information wired from VIs previously called. Use this information to decide if any functionality should be bypassed in the event of errors from other VIs.
- Right-click the **error in** control on the front panel and select **Explain Error** or **Explain Warning** from the shortcut menu for more information about the error.
- status is TRUE (X) if an error occurred or FALSE (checkmark) to indicate a warning or that no error occurred.
- Right-click the **error in** control on the front panel and select **Explain Error** or **Explain Warning** from the shortcut menu for more information about the error.
- code is the error or warning code.
- Right-click the **error in** control on the front panel and select **Explain Error** or **Explain Warning** from the shortcut menu for more information about the error.
- source describes the origin of the error or warning.
- Right-click the **error in** control on the front panel and select **Explain Error** or **Explain Warning** from the shortcut menu for more information about the error.

- DAQQueues
 - SignalList
 - CAS
 - CycAvg
 - Queues
 - CAS
 - CycAvg
 - scaled units
 - scaled units





- APC_Pcyl_Diag.vi
- C:\LabVIEW PROJECT\MONARCH\support\APC_Pcyl_Diag.vi
- APC_CombustionMetrics.cti
- C:\LabVIEW PROJECT\MONARCH\controls\APC_CombustionMetrics.cti
- APC_PcylDiag.cti
- C:\LabVIEW PROJECT\MONARCH\controls\APC_PcylDiag.cti
- APC_9049_HRL_IMEP.vi
- C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_IMEP.vi
- APC_9049_HRL_apparentHRL.vi
- C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_apparentHRL.vi
- APC_9049_HRL_pegging.vi
- C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_pegging.vi
- APC_9049_HRL_ZeroPhaseFilter.vi
- C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_ZeroPhaseFilter.vi
- APC_9049_HRL_MAPO.vi
- C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_MAPO.vi
- APC_9049_HRL_volume.vi
- C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_volume.vi

APC_Pcyl_Diag.vi

CombustionAnalysisCluster [3] [1] CombustionAnalysisClusterOut
Expected IMEP [2] [0] WarningsAndErrors

Load INI on startup

Expected IMEP

CombustionAnalysisCluster

CombustionAnalysisClusterOut

WarningsAndErrors

samples for running IMEP std

MaxDevFromExpectedIMEPWarning [bar]

MaxDevFromExpectedIMEPErrror [bar]

MaxDevFromSelfAvgWarning [bar]

MaxDevFromAvgError [bar]

MaxDevFromAvgWarning [bar]

MaxDevFromSelfAvgError [bar]

MaxIMEPstdWarning [bar]

MaxIMEPstdError [bar]

MAPOmaxWarning [bar/CAD]

MAPOmaxError [bar/CAD]

CA50maxWarning [CADATDC]

CA50maxError [CADATDC]

MaxPCylMaxWarning [bar]

MaxPCylMaxError [bar]

IMEPn [bar]

IMEPg [bar]

MAPO [bar/CAD]

IMEPn [bar]

IMEPg [bar]

MAPO [bar/CAD]

maxQ [J]

CA10 [CADATDC]

CA50 [CADATDC]

CA90 [CADATDC]

Pmax [bar]

IMEPstd [bar]

CA03 [CADATDC]

CA97 [CADATDC]

CA10CA90 [CADATDC]

maxdpdCAD [bar/CAD]

kappa_comp [-]

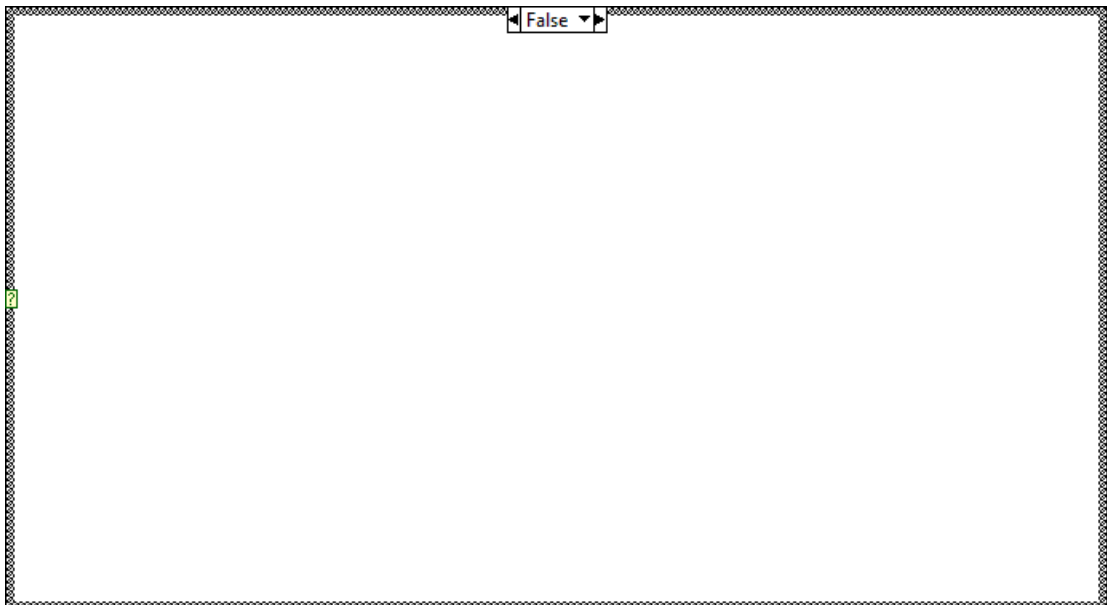
kappa_exp [-]

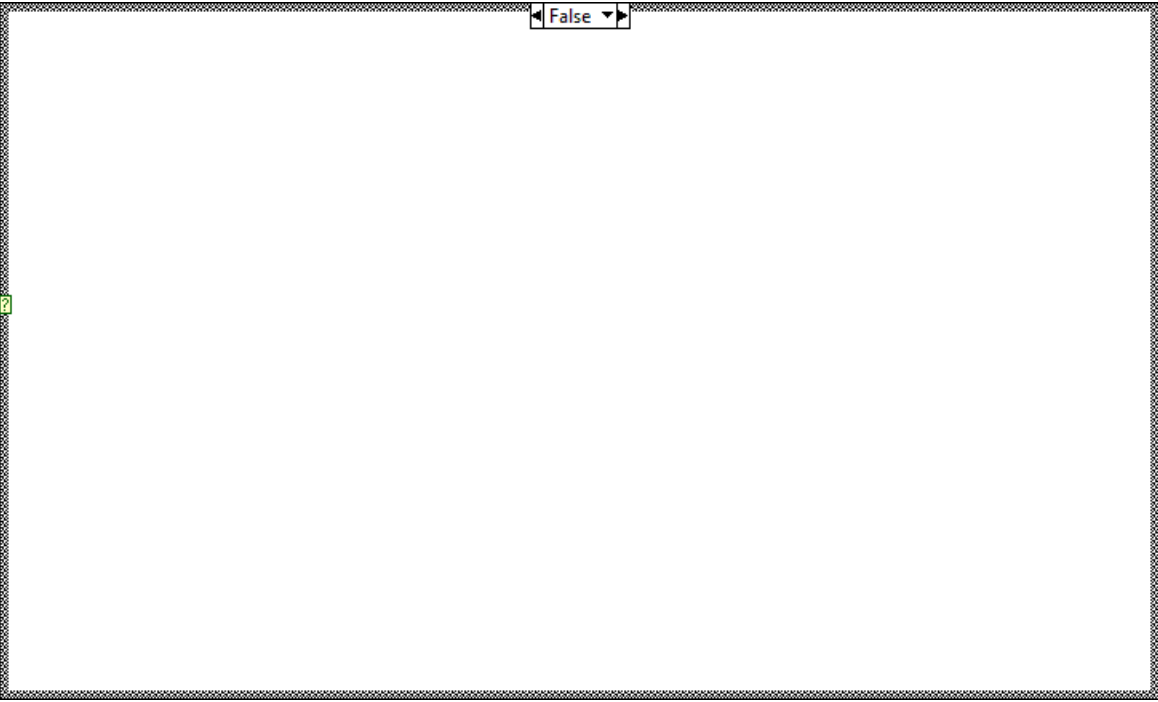
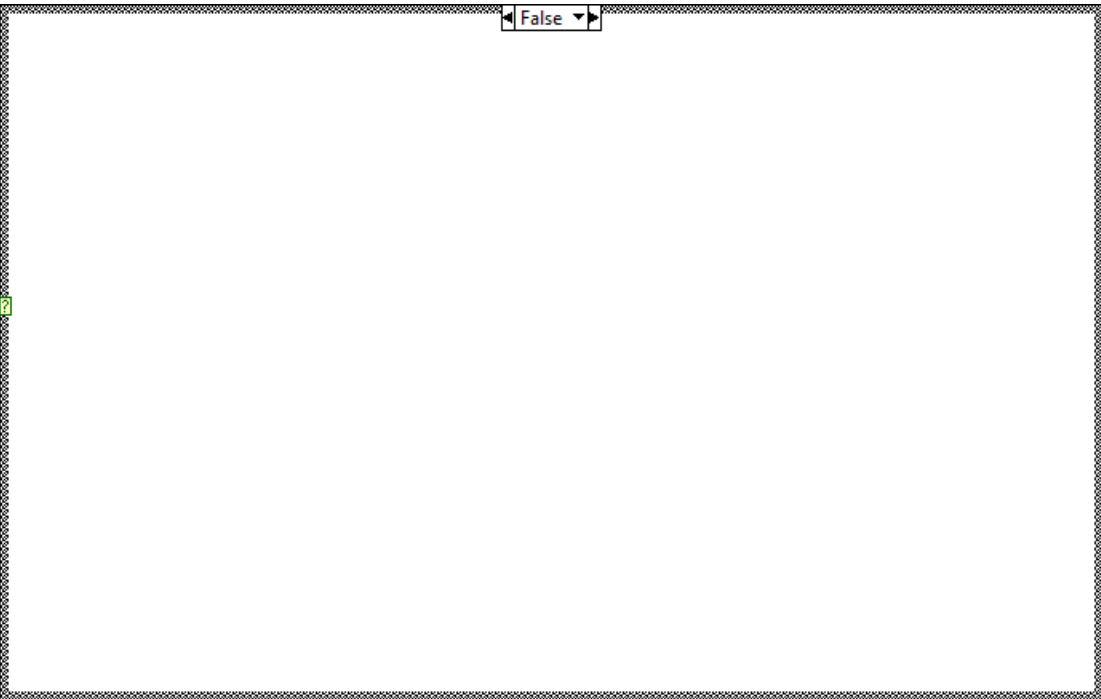
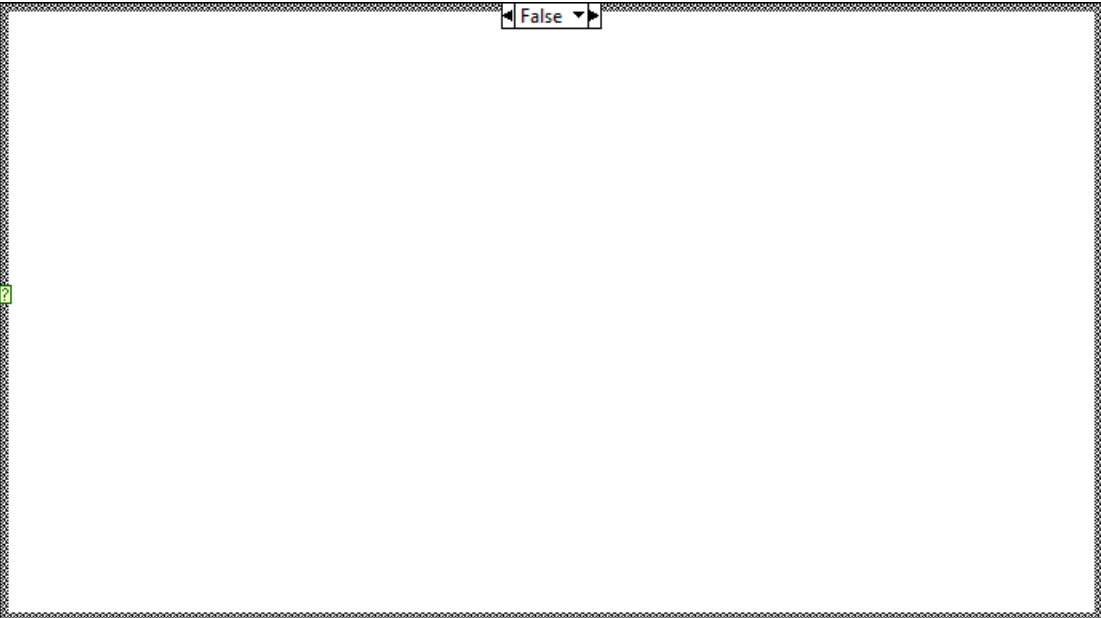
PMEP [bar]

Ri [-]

CombustionAnalysisCluster

- IMEPn [bar]
- IMEPn [bar]
- IMEPg [bar]
- IMEPg [bar]
- MAPO [bar/CAD]
- maxQ [J]
- maxQ [J]
- CA10 [CADATDC]
- CA10
- CA50 [CADATDC]
- CA50
- CA90 [CADATDC]
- CA50
- Pmax [bar]
- CA50
- IMEPstd [bar]
- CA50
- CA03 [CADATDC]
- CA10
- CA97 [CADATDC]
- CA10
- CA10CA90 [CADATDC]
- CA10
- maxdpdCAD [bar/CAD]
- CA10

CombustionAnalysisClusterOut



- NI_PtByPt.lib:Standard Deviation PtByPt.vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\ptbypt(Probability & Statistics.lib\Standard Deviation PtByPt.vi
- APC_9049_LoadWarningConfig.vi**
C:\LabVIEW PROJECT\MONARCH\APC_9049_LoadWarningConfig.vi
- APC_CombustionMetrics.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_CombustionMetrics.ctl
- APC_PcylDiag.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_PcylDiag.ctl
- NI_AALBase.lib:Mean (DBL).vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\bascanly.lib\Mean (DBL).vi
- NI_AALBase.lib:Mean.vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\bascanly.lib\Mean.vi
- APC_CylDagnosticsCluster.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_CylDagnosticsCluster.ctl

APC_9049_LoadWarningConfig.vi

Load? [3]

CylinderDiagnosticsCluster [2]

[0] CylinderDiagnosticsCluster 2

error out

status code

source

CylinderDiagnosticsCluster

samples for running IMEP std

100

MaxDevFromExpectedIMEPError [bar]

2

MaxDevFromExpectedIMEPWarning [bar]

2

MaxDevFromAvgError [bar]

2

MaxDevFromSelfAvgWarning [bar]

2

MaxDevFromSelfAvgError [bar]

2

MaxDevFromAvgWarning [bar]

2

MaxIMEPstdWarning [bar]

0.5

MaxIMEPstdError [bar]

0.5

MAPOmaxError [bar/CAD]

1

MAPOmaxWarning [bar/CAD]

1

CA50maxError [CADATDC]

15

CA50maxWarning [CADATDC]

15

MaxPCylIMaxError [bar]

120

MaxPCylIMaxWarning [bar]

120

CylinderDiagnosticsCluster 2

samples for running IMEP std

100

MaxDevFromExpectedIMEPError [bar]

2

MaxDevFromExpectedIMEPWarning [bar]

2

MaxDevFromAvgError [bar]

2

MaxDevFromSelfAvgWarning [bar]

2

MaxDevFromSelfAvgError [bar]

2

MaxDevFromAvgWarning [bar]

2

MaxIMEPstdWarning [bar]

0.5

MaxIMEPstdError [bar]

0.5

MAPOmaxError [bar/CAD]

1

MAPOmaxWarning [bar/CAD]

1

CA50maxError [CADATDC]

15

CA50maxWarning [CADATDC]

15

MaxPCylIMaxError [bar]

120

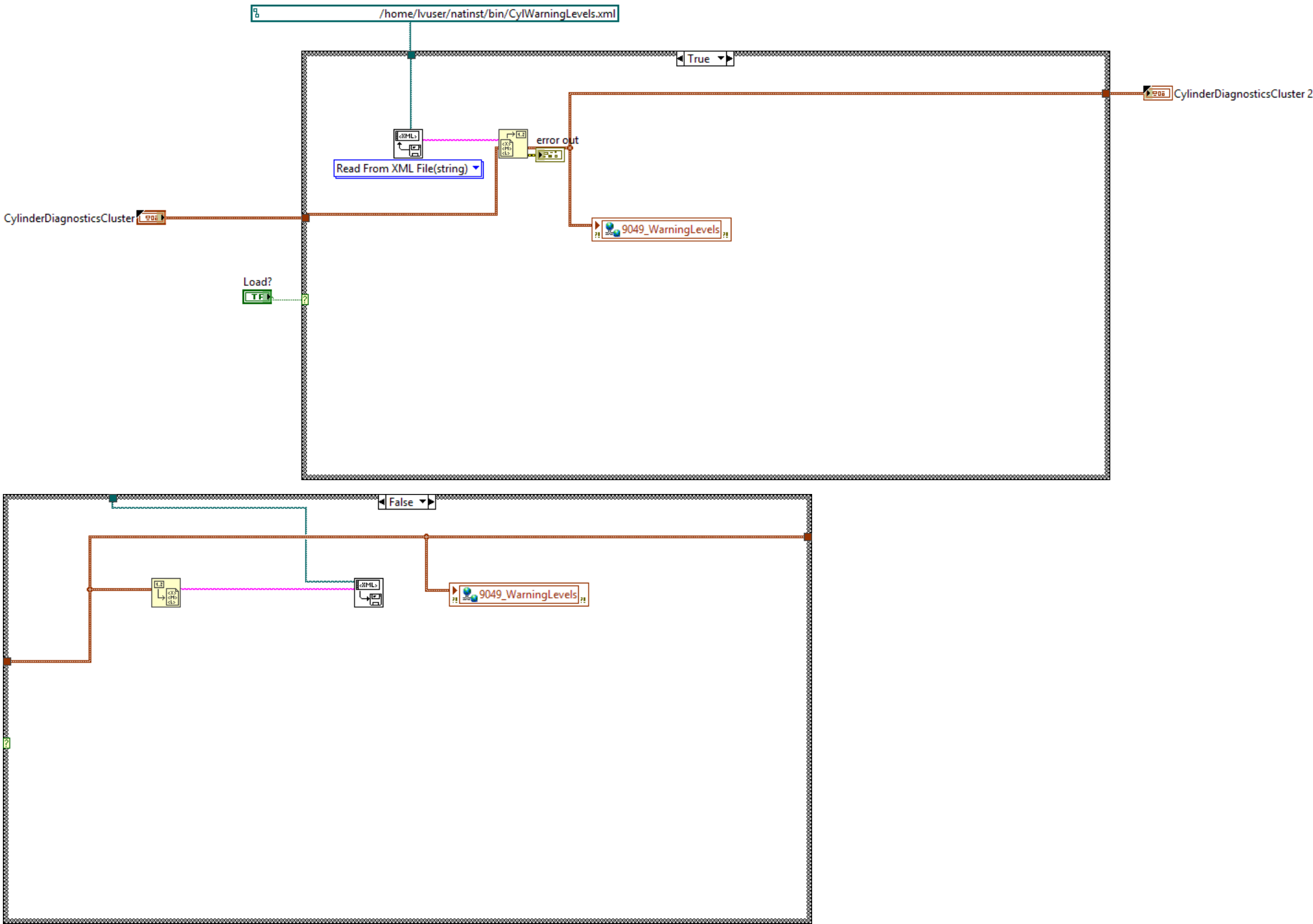
MaxPCylIMaxWarning [bar]

120

Load?

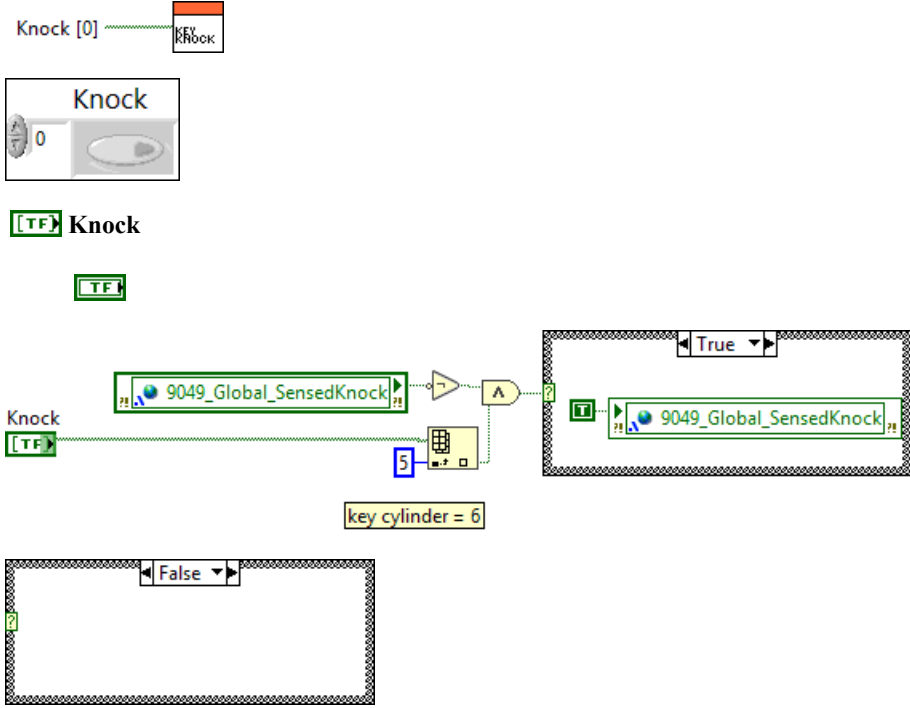
- Load?**
- CylinderDiagnosticsCluster**
 - samples for running IMEP std
 - MaxDevFromExpectedIMEPError [bar]
 - MaxDevFromExpectedIMEPWarning [bar]
 - MaxDevFromAvgError [bar]
 - MaxDevFromSelfAvgWarning [bar]
 - MaxDevFromSelfAvgError [bar]
 - MaxDevFromAvgWarning [bar]
 - MaxIMEPstdWarning [bar]
 - MaxIMEPstdError [bar]
 - MAPOmaxError [bar/CAD]
 - MAPOmaxWarning [bar/CAD]
 - CA50maxError [CADATDC]
 - CA50maxWarning [CADATDC]
 - MaxPCylIMaxError [bar]
 - MaxPCylIMaxWarning [bar]

- CylinderDiagnosticsCluster 2**
 - samples for running IMEP std
 - MaxDevFromExpectedIMEPError [bar]
 - MaxDevFromExpectedIMEPWarning [bar]
 - MaxDevFromAvgError [bar]
 - MaxDevFromSelfAvgWarning [bar]
 - MaxDevFromSelfAvgError [bar]
 - MaxDevFromAvgWarning [bar]
 - MaxIMEPstdWarning [bar]
 - MaxIMEPstdError [bar]
 - MAPOmaxError [bar/CAD]
 - MAPOmaxWarning [bar/CAD]
 - CA50maxError [CADATDC]
 - CA50maxWarning [CADATDC]
 - MaxPCylIMaxError [bar]
 - MaxPCylIMaxWarning [bar]



- Read From XML File(string).vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Utility\xml.lib\Read From XML File(string).vi
- Read From XML File.vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Utility\xml.lib\Read From XML File.vi
- Write to XML File(string).vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Utility\xml.lib\Write to XML File(string).vi
- Write to XML File.vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Utility\xml.lib\Write to XML File.vi
- APC_CylDagnosticsCluster.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_CylDagnosticsCluster.ctl

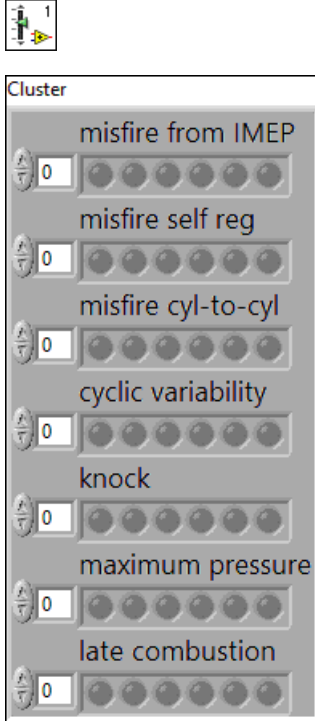
APC_9049_keyKnock.vi



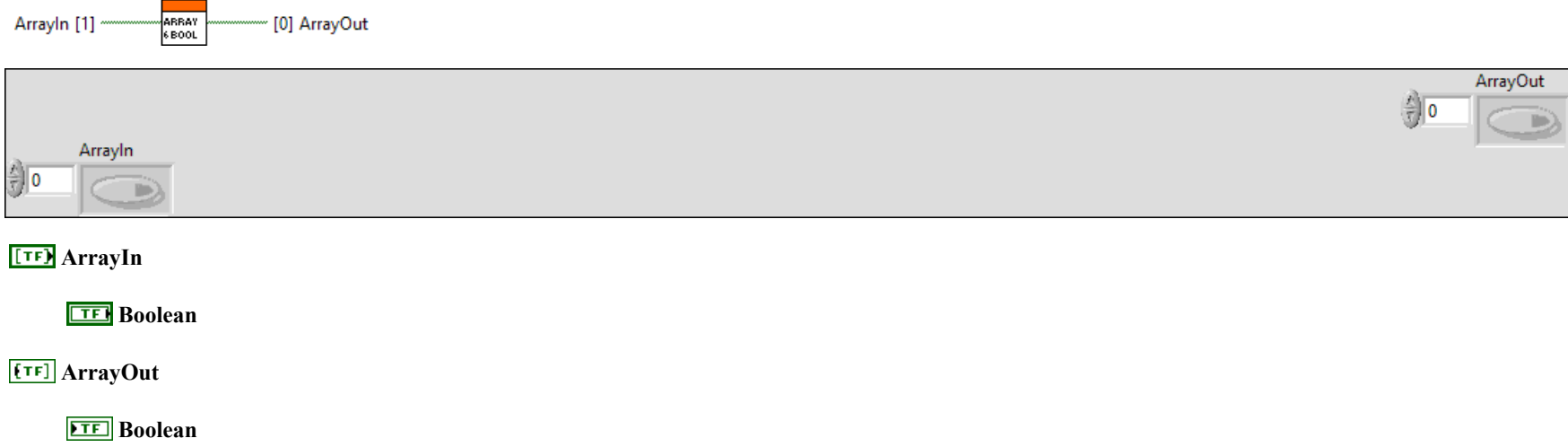
APC_CombustionMetrics.ctl

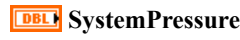
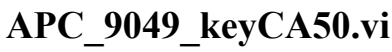
IMEPn [bar]					
NaN	NaN	NaN	NaN	NaN	NaN
IMEPg [bar]					
NaN	NaN	NaN	NaN	NaN	NaN
MAPO [bar/CAD]					
NaN	NaN	NaN	NaN	NaN	NaN
maxQ [J]					
NaN	NaN	NaN	NaN	NaN	NaN
CA10 [CADATDC]					
NaN	NaN	NaN	NaN	NaN	NaN
CA50 [CADATDC]					
NaN	NaN	NaN	NaN	NaN	NaN
CA90 [CADATDC]					
NaN	NaN	NaN	NaN	NaN	NaN
Pmax [bar]					
NaN	NaN	NaN	NaN	NaN	NaN
IMEPstd [bar]					
NaN	NaN	NaN	NaN	NaN	NaN
CA03 [CADATDC]					
NaN	NaN	NaN	NaN	NaN	NaN
CA97 [CADATDC]					
NaN	NaN	NaN	NaN	NaN	NaN
CA10CA90 [CADATDC]					
NaN	NaN	NaN	NaN	NaN	NaN
maxdpdCAD [bar/CAD]					
NaN	NaN	NaN	NaN	NaN	NaN
kappa_comp [-]					
NaN	NaN	NaN	NaN	NaN	NaN
kappa_exp [-]					
NaN	NaN	NaN	NaN	NaN	NaN
PMEP [bar]					
NaN	NaN	NaN	NaN	NaN	NaN
RI [-]					
NaN	NaN	NaN	NaN	NaN	NaN

APC_PcylDiag.ctl



APC_9056_ForceArraySize6b.vi





mean is the mean, or average, of the values in the input sequence **X**

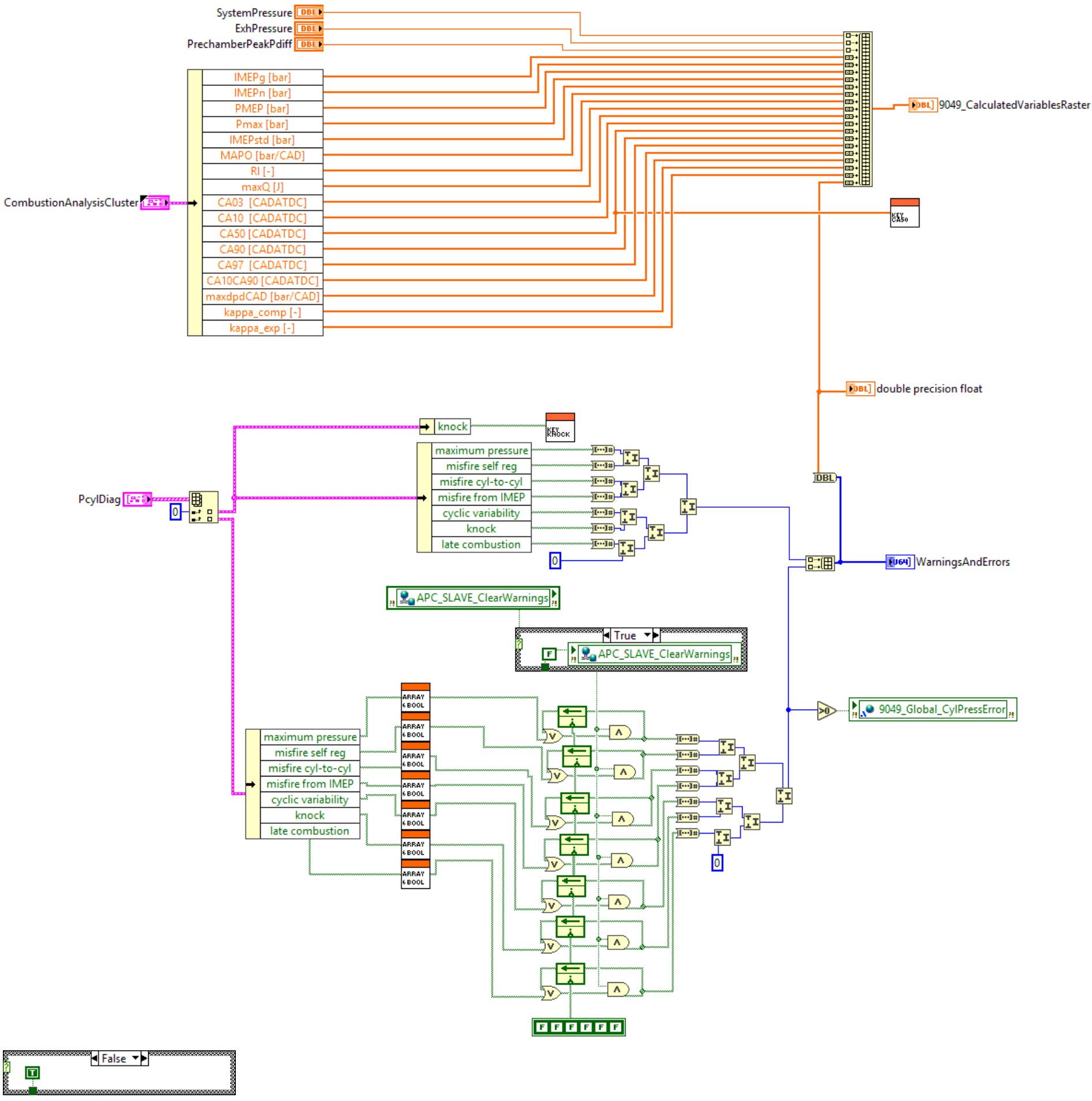


mean is the mean, or average, of the values in the input sequence **X**



mean is the mean, or average, of the values in the input sequence **X**



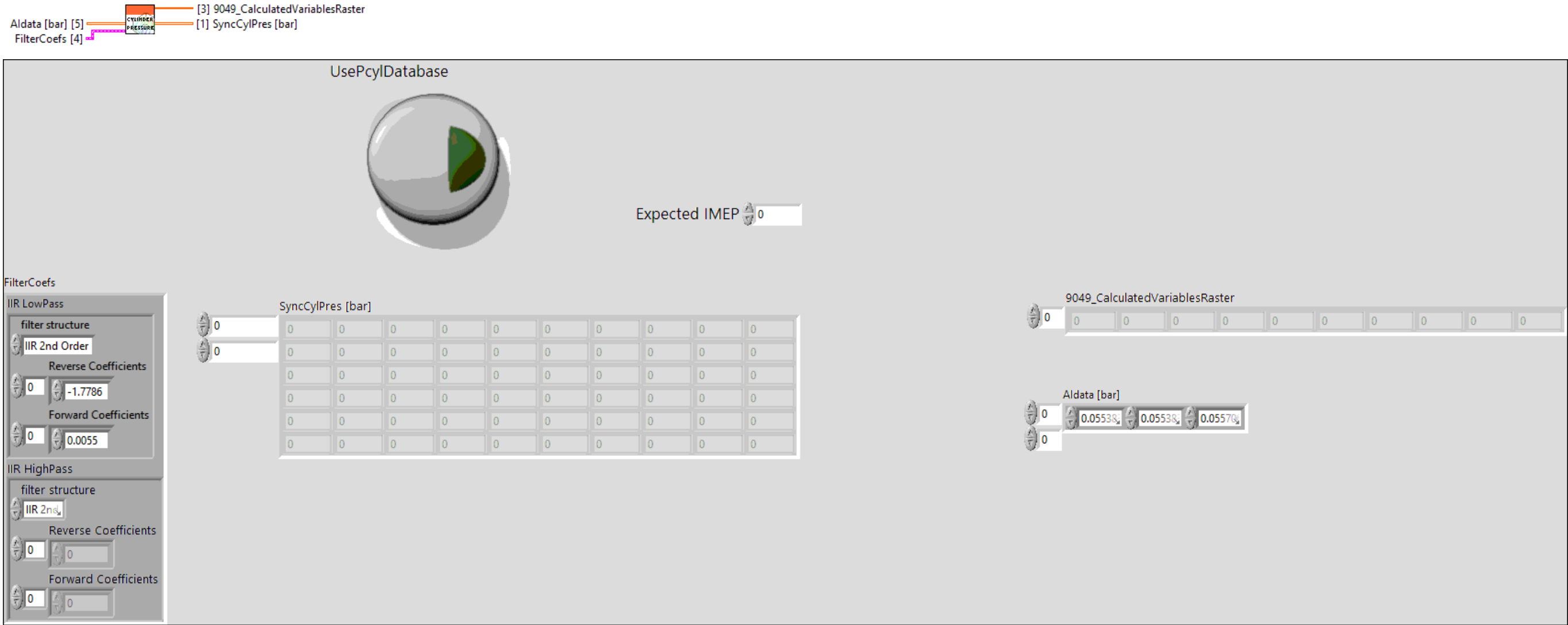


- APC_9049_keyKnock.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_keyKnock.vi
- APC_CombustionMetrics.ctl
C:\LabVIEW PROJECT\MONARCH\controls\APC_CombustionMetrics.ctl
- APC_PcyIDiag.ctl
C:\LabVIEW PROJECT\MONARCH\controls\APC_PcyIDiag.ctl
- APC_9056_ForceArraySize6b.vi
C:\LabVIEW PROJECT\MONARCH\APC_9056_ForceArraySize6b.vi
- APC_9049_keyCA50.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_keyCA50.vi

APC_9049_PressureAnalytics.vi

This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.



Aldata [bar]

element

FilterCofes

IIR LowPass

IIR Filter Cluster contains the cascaded form of IIR filter coefficients.

filter structure

filter structure indicates either IIR second-order or IIR fourth-order filter stages.

Reverse Coefficients

Reverse Coefficients is the reverse coefficients of the IIR cascade filter.

Reverse Coefficients is the reverse coefficients of the IIR cascade filter.

Forward Coefficients

Forward Coefficients is the forward coefficients of the IIR cascade filter.

Forward Coefficients is the forward coefficients of the IIR cascade filter.

IIR HighPass

filter structure

Reverse Coefficients

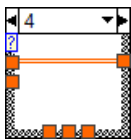
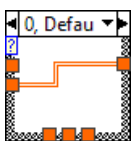
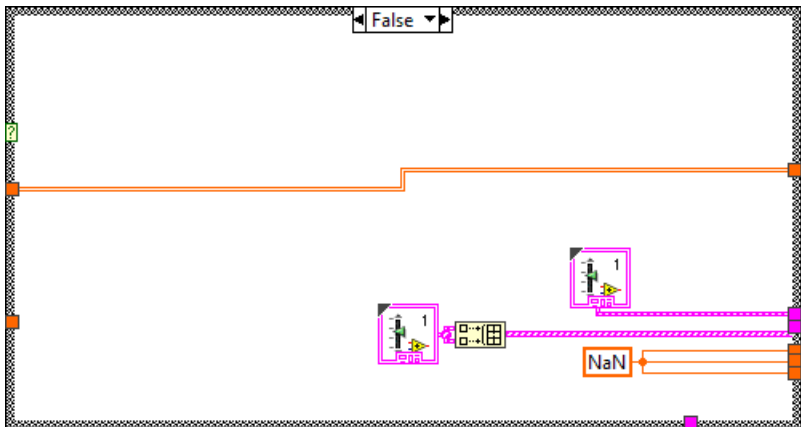
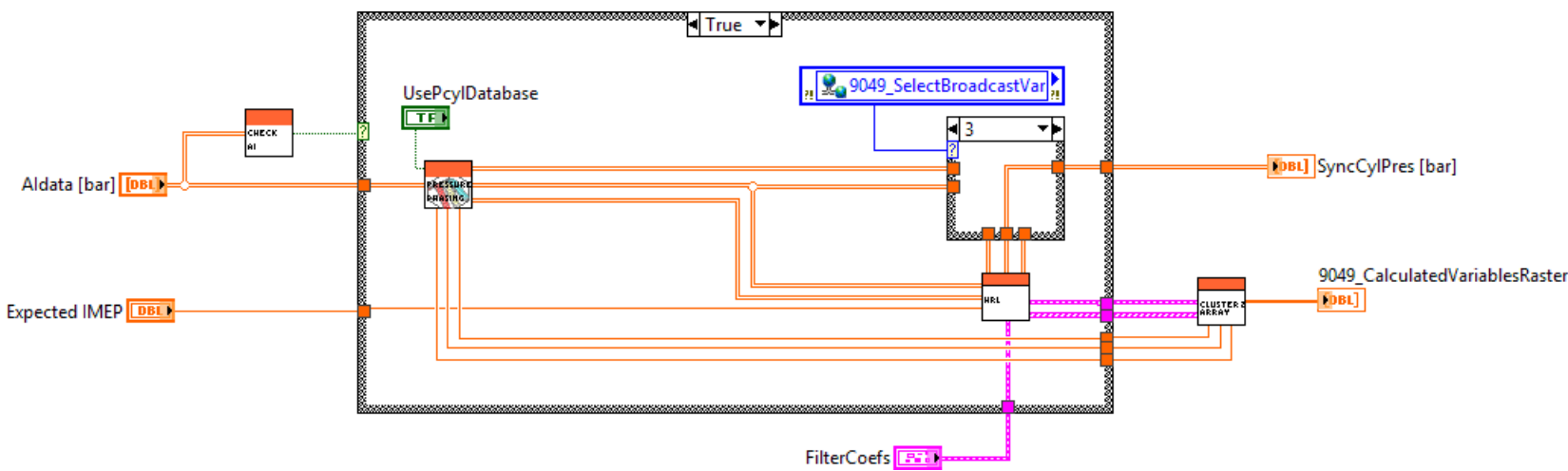
Forward Coefficients

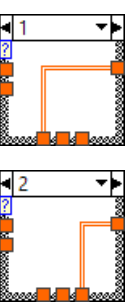
SyncCylIPres [bar]

element

9049_CalculatedVariablesRaster

mean



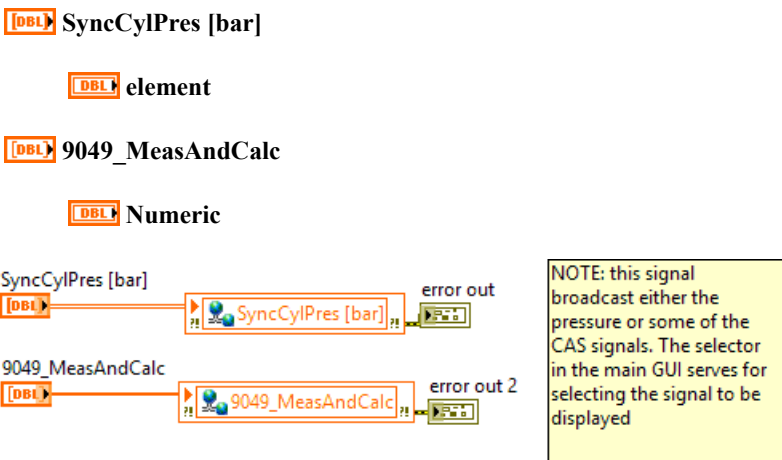
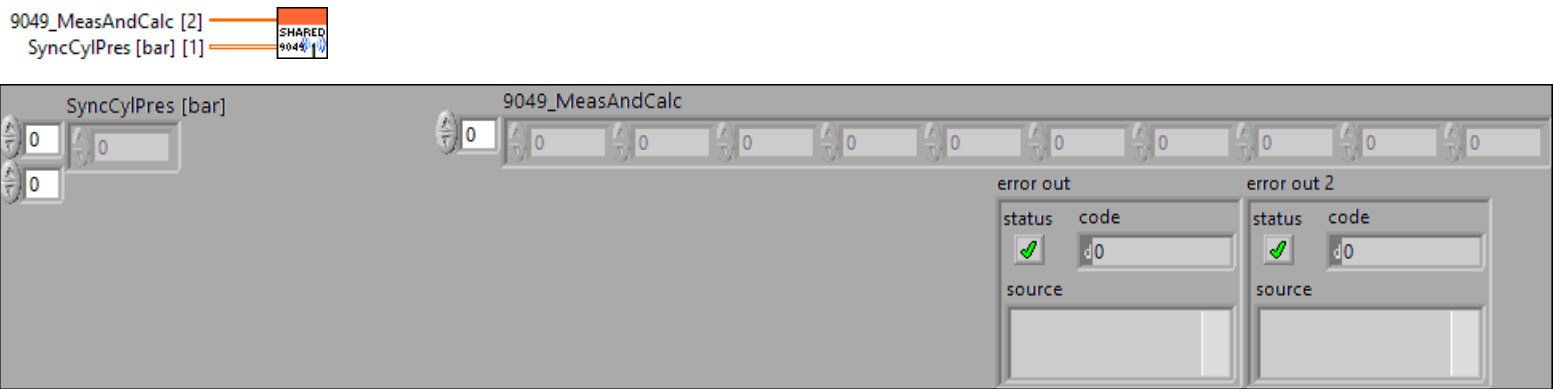


- APC_HRL.vi**
C:\LabVIEW PROJECT\MONARCH\support\APC_HRL.vi
- APC_CombustionMetrics.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_CombustionMetrics.ctl
- APC_PeylDiag.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_PeylDiag.ctl
- APC_9049_CombCluster2Array.vi**
C:\LabVIEW PROJECT\MONARCH\support\APC_9049_CombCluster2Array.vi
- APC_9049_PPhaseCorrection.vi**
C:\LabVIEW PROJECT\MONARCH\APC_9049_PPhaseCorrection.vi
- APC_9049_checkAI.vi**
C:\LabVIEW PROJECT\MONARCH\APC_9049_checkAI.vi

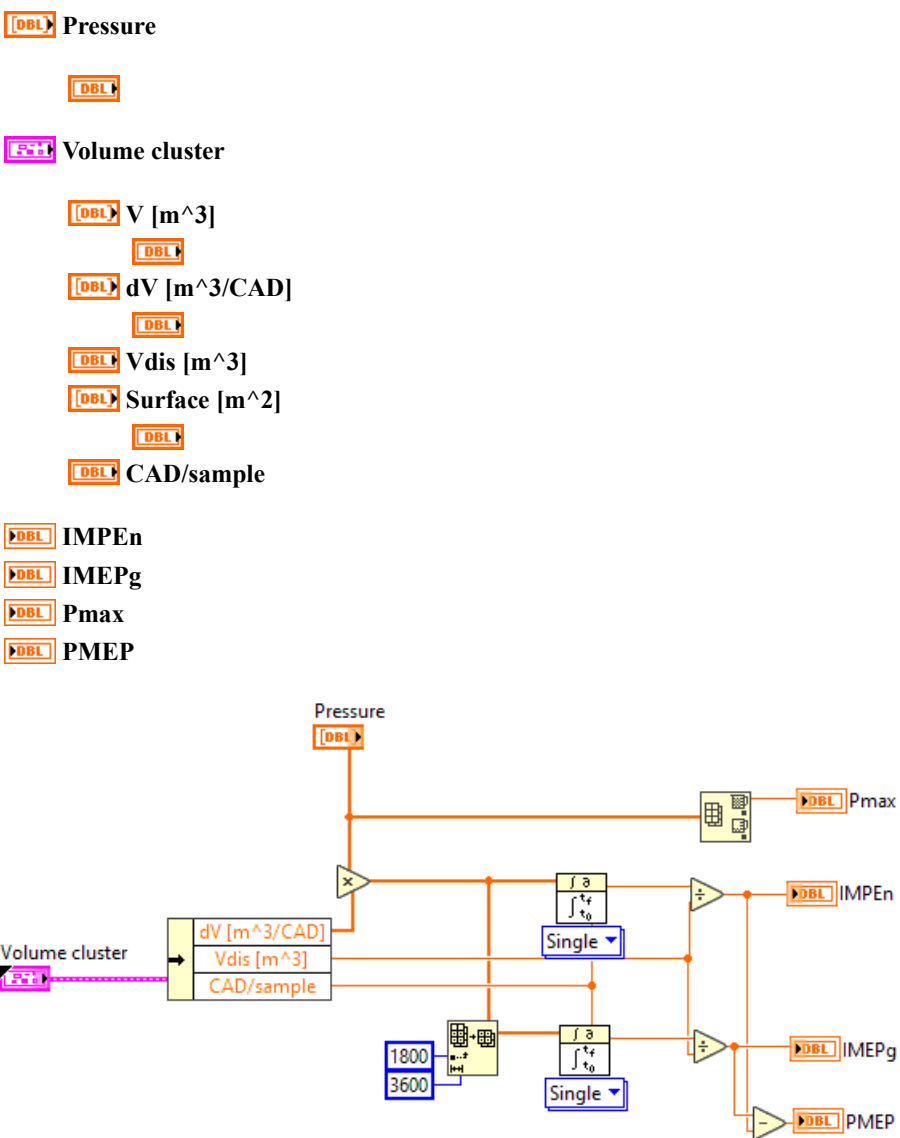
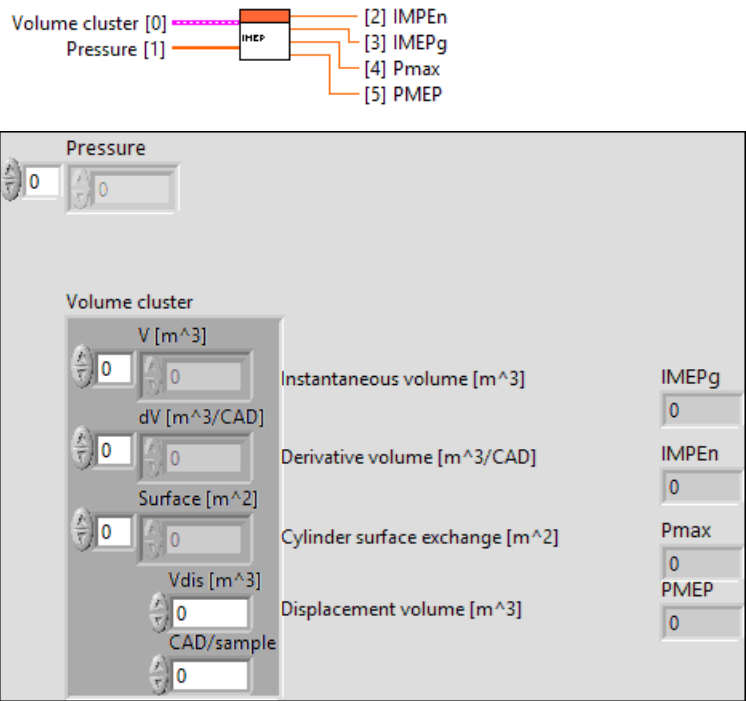
APC_9049_SharedVarBroadcast.vi

This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.

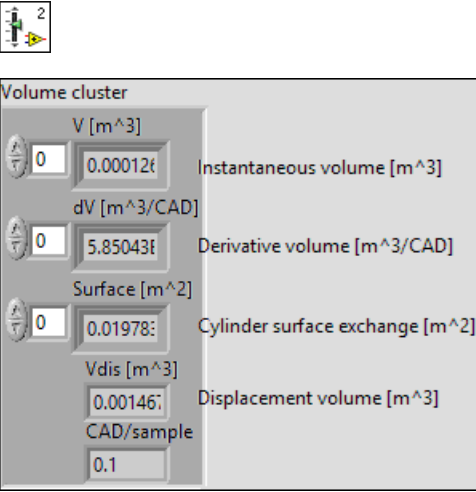


APC_9049_HRL_IMEP.vi

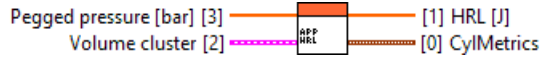


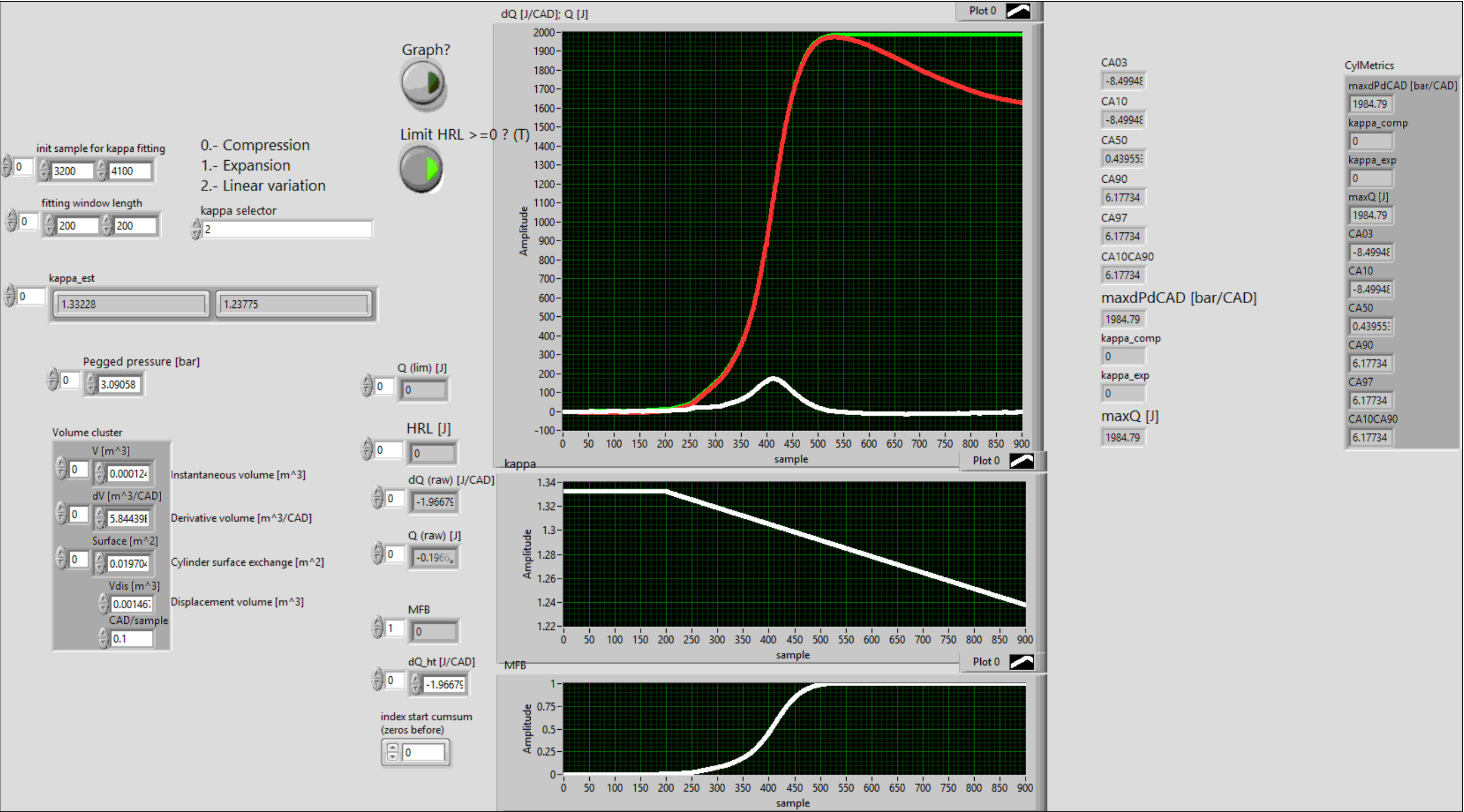
- NI_AALPro.lib:1D Numeric Integration.vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\8numeric.llb\1D Numeric Integration.vi
- NI_AALPro.lib:Numeric Integration.vi**
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\8numeric.llb\Numeric Integration.vi
- APC_volume.ctl**
C:\LabVIEW PROJECT\MONARCH\controls\APC_volume.ctl

APC_volume.ctl



APC_9049_HRL_apparentHRL.vi





Pegged pressure [bar]

Volume cluster

V [m³]

dV [m³/CAD]

Vdis [m³]

Surface [m²]

CAD/sample

HRL [J]

HRL [J]

CylMetrics

maxdPdCAD [bar/CAD]

kappa_comp

kappa_exp

maxQ [J]

CA03

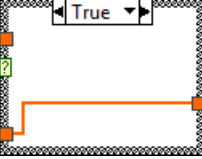
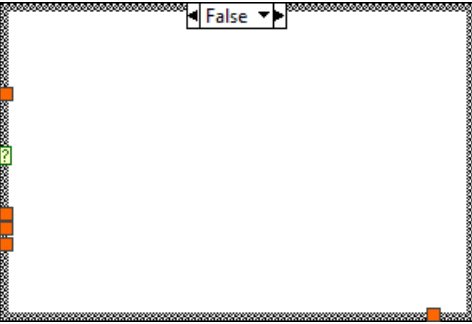
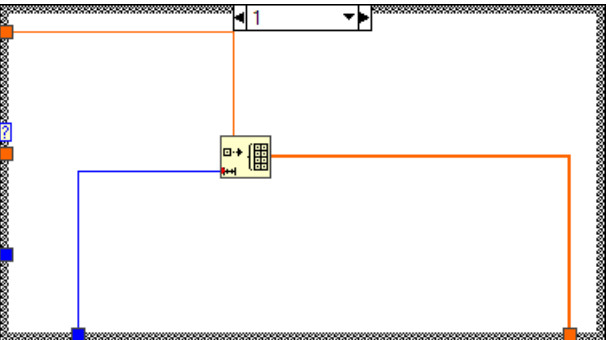
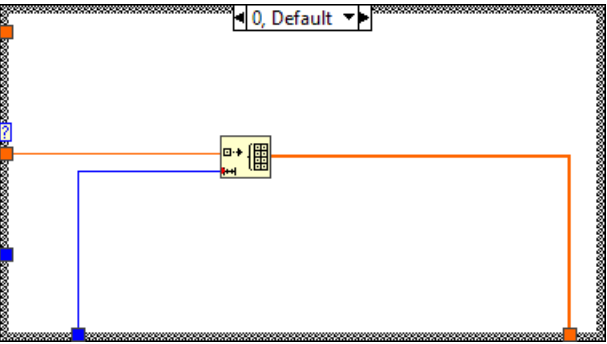
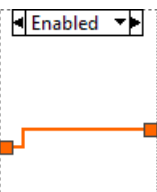
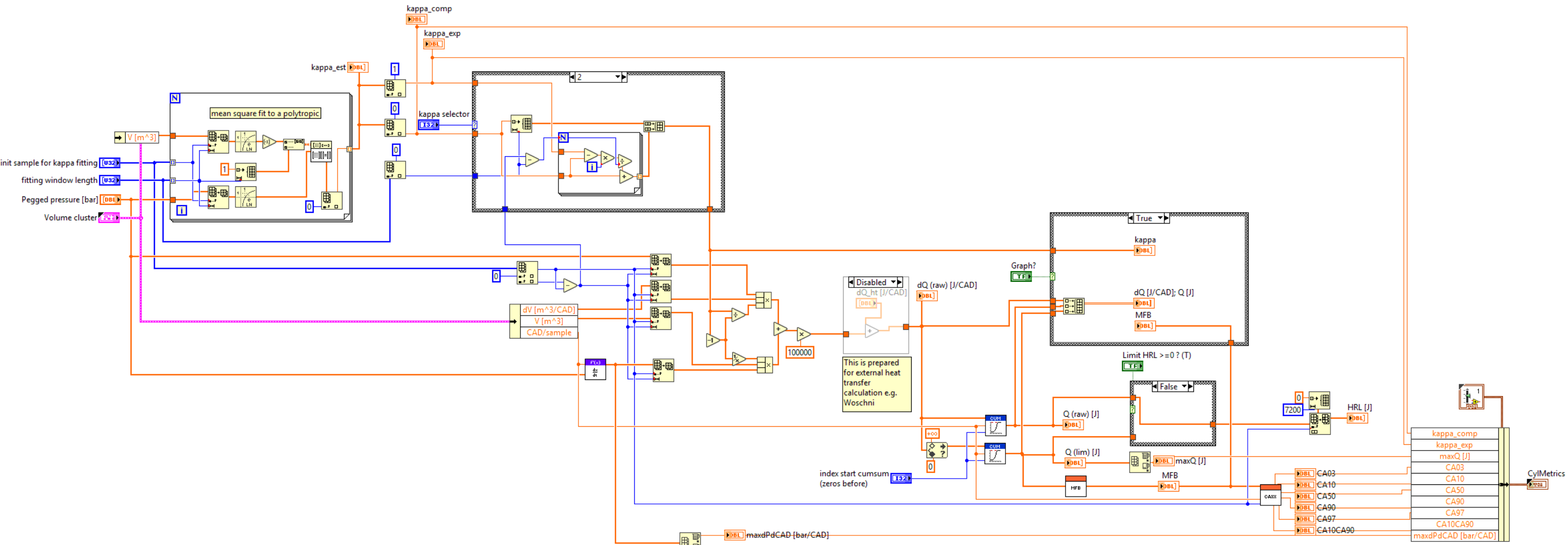
CA10

CA50

CA90

CA97

CA10CA90



APC_volume.ctf

C:\LabVIEW PROJECT\MONARCH\controls\APC_volume.ctf

APC_9049_CA.vi

C:\LabVIEW PROJECT\MONARCH\APC_9049_CA.vi

APC_9049_MFB.vi

C:\LabVIEW PROJECT\MONARCH\APC_9049_MFB.vi

cumsum_dQ.vi

C:\LabVIEW PROJECT\MONARCH\cumsum_dQ.vi

NI_AALPro.Ivlib:Solve Linear Eqs (Gen, single right hand).vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\7finalg.llb\Solve Linear Eqs (Gen, single right hand).vi

APC_CAD_derivative.vi

C:\LabVIEW PROJECT\MONARCH\support\APC_CAD_derivative.vi

NI_AALPro.Ivlib:Solve Linear Equations (single right hand).vi

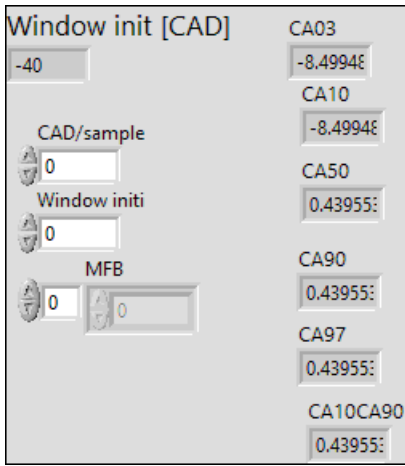
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\7finalg.llb\Solve Linear Equations (single right hand).vi

NI_AALPro.Ivlib:Solve Linear Equations.vi

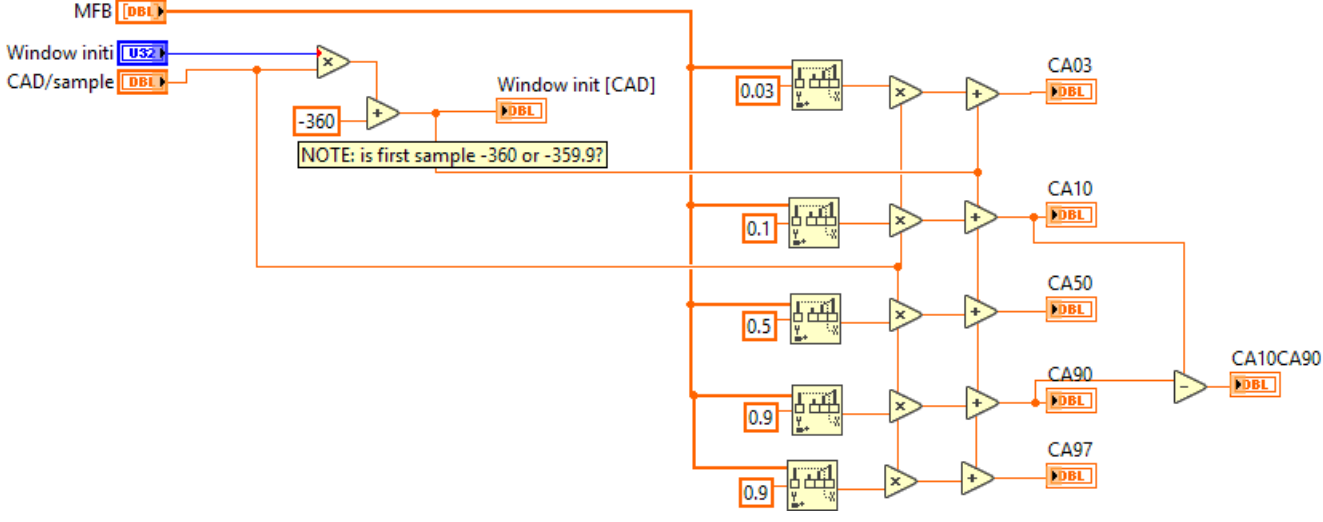
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\7finalg.llb\Solve Linear Equations.vi

APC_CylMetrics.ctf

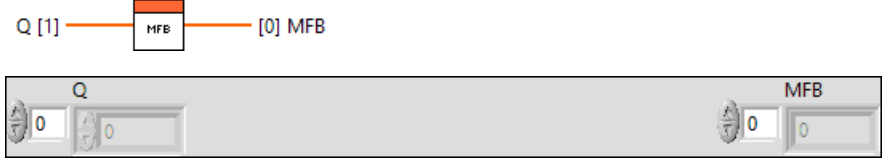
APC_9049_CA.vi



-  CA10
-  CA50
-  CA90
-  CA03
-  CA97
- CA10CA90

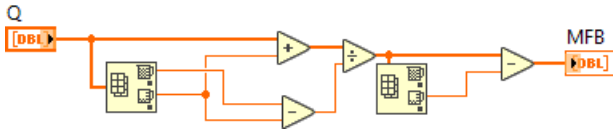


APC_9049_MFB.vi

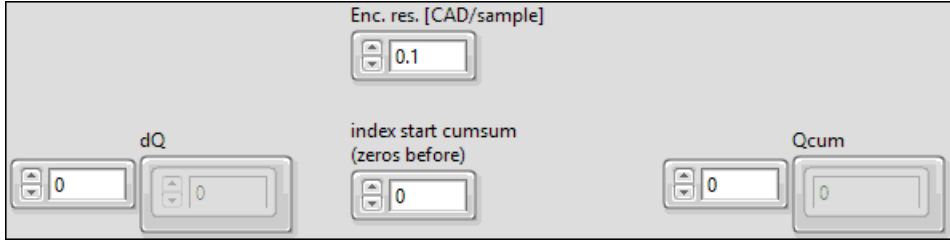
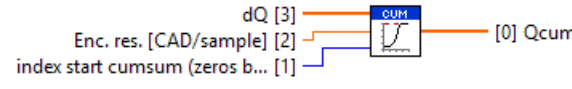


Q

MFB



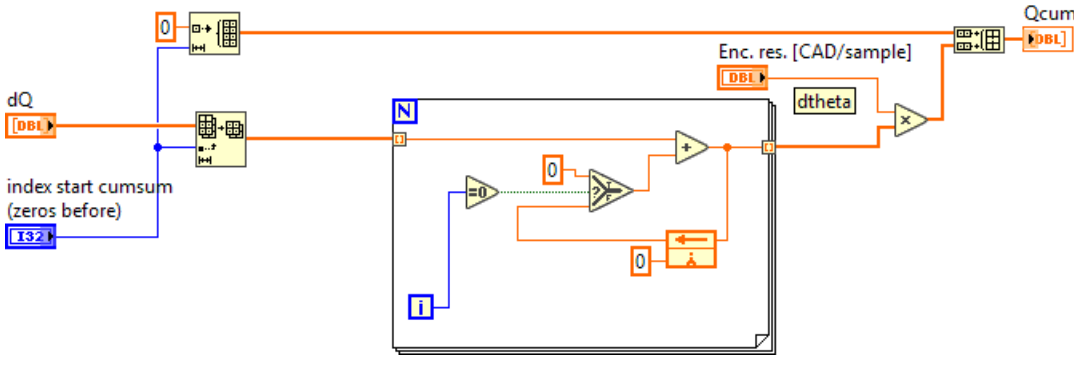
cumsum_dQ.vi



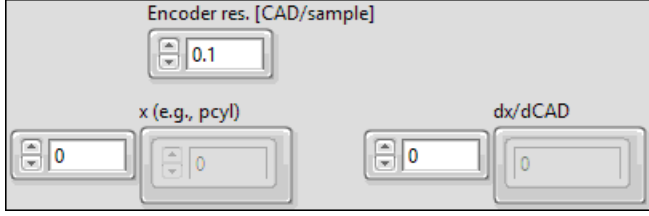
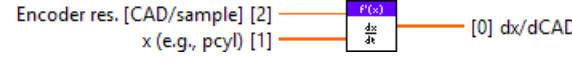
[132] index

[DBL] dQ

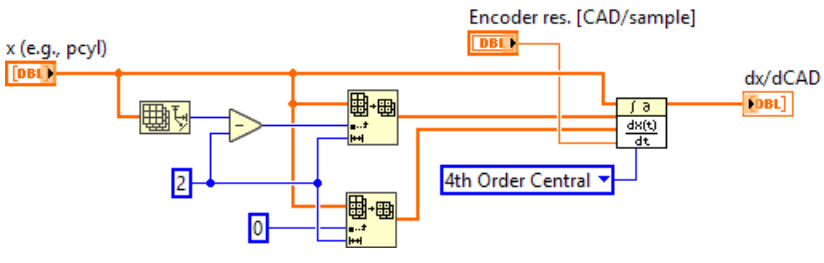
-  Numeric Control
- Enc. res. [CAD/sample]
-  Qcum
-  element



APC_CAD_derivative.vi



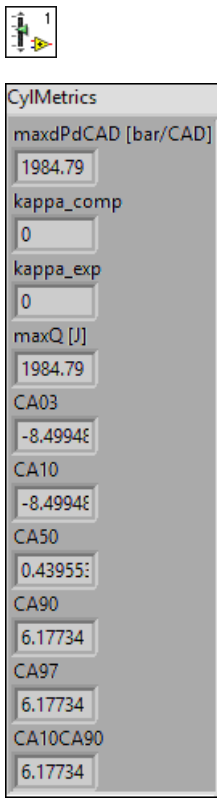
- DSL x (e.g., pcyll)
- DSL Numeric
- DSL Encoder res. [
- DSL dx/dCAD
- DSL Numeric



NI_AALPro.lvlib:Derivative x(t).vi

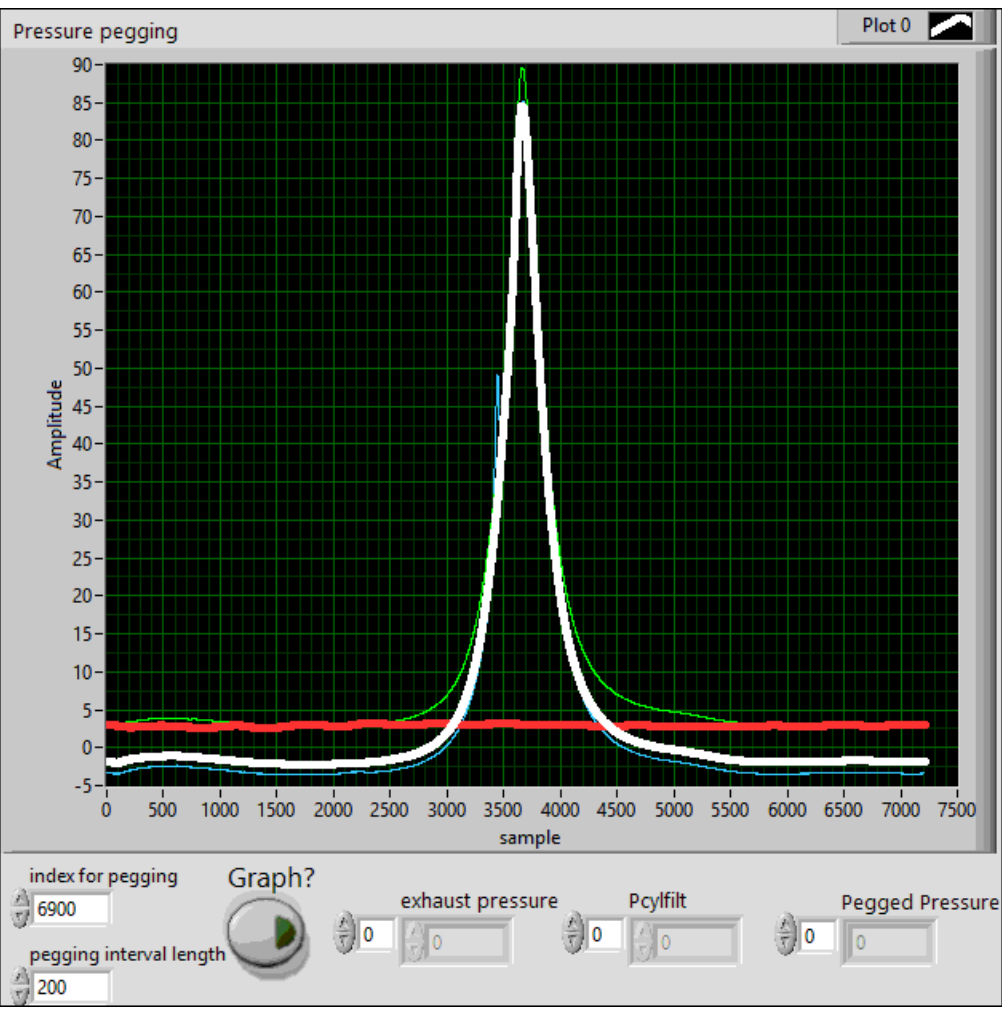
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\2dsp.llb\Derivative x(t).vi

APC_CylMetrics.ctl



APC_9049_HRL_pegging.vi





exhaust pressure

Numeric

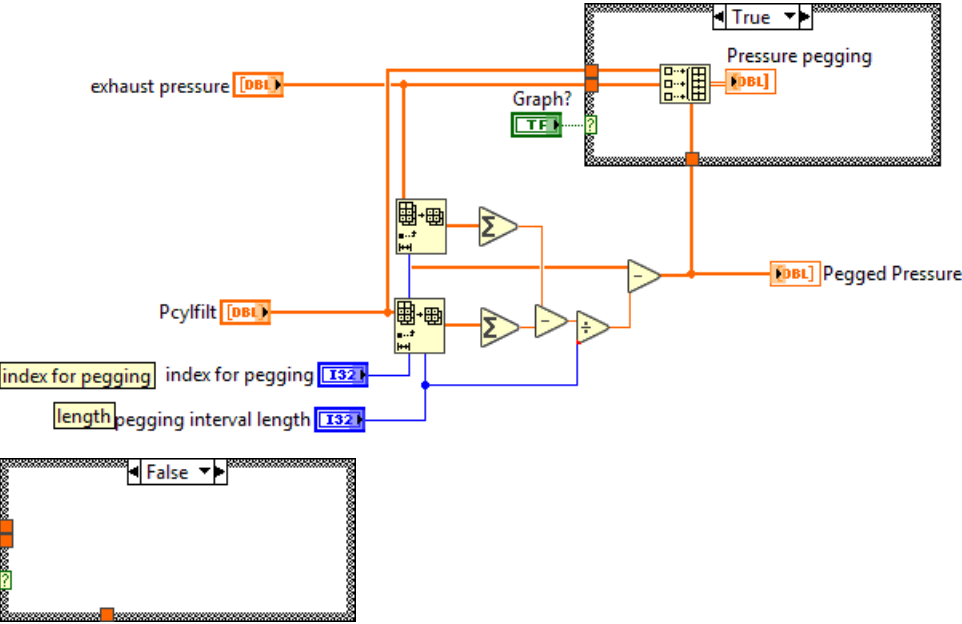
Pcylfilt

DBL

Pressure pegging

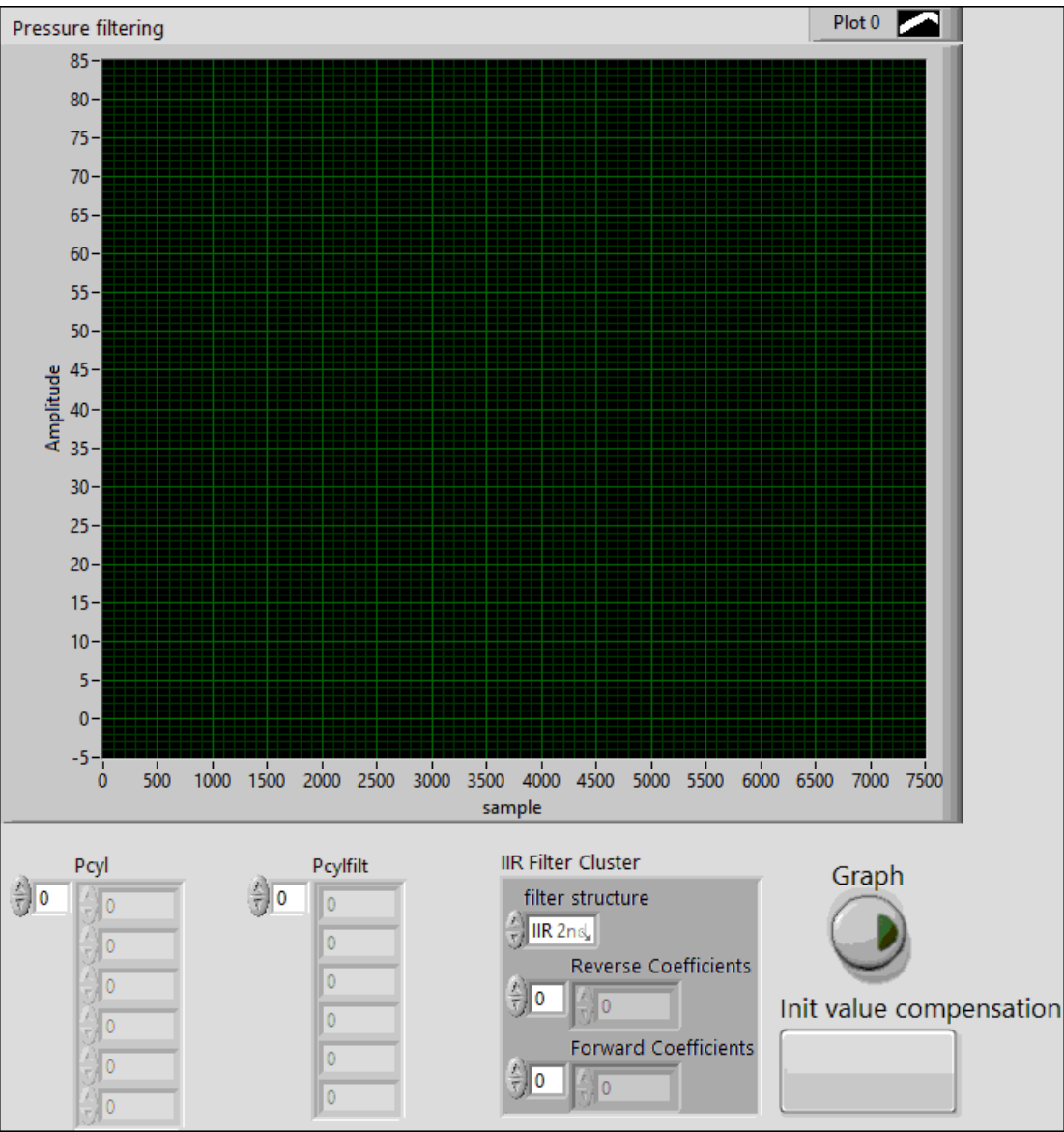
Pegged Pressure

DBL



APC_9049_HRL_ZeroPhaseFilter.vi

Pcyl [4] [1] Pressure filtering
IIR Filter Cluster [3] [0] Pcylfilt
Init value compensation [2]



Pcyl

Numeric

IIR Filter Cluster

filter structure

Reverse Coefficients

DBL

Forward Coefficients

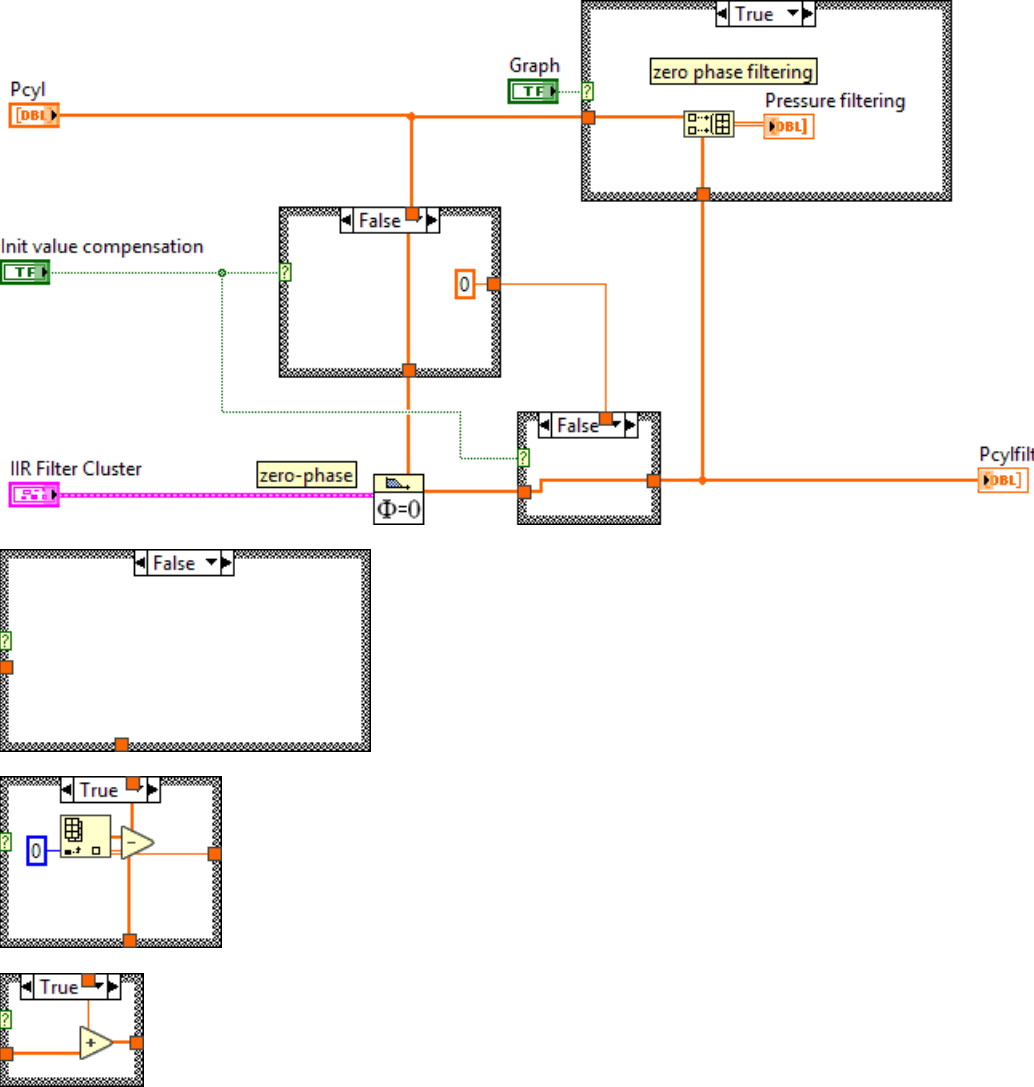
DBL

Init value compensation

Pressure filtering

Pcylfilt

DBL



NI_AALBase.lib:Zero Phase Filter (Cascade, DBL).vi

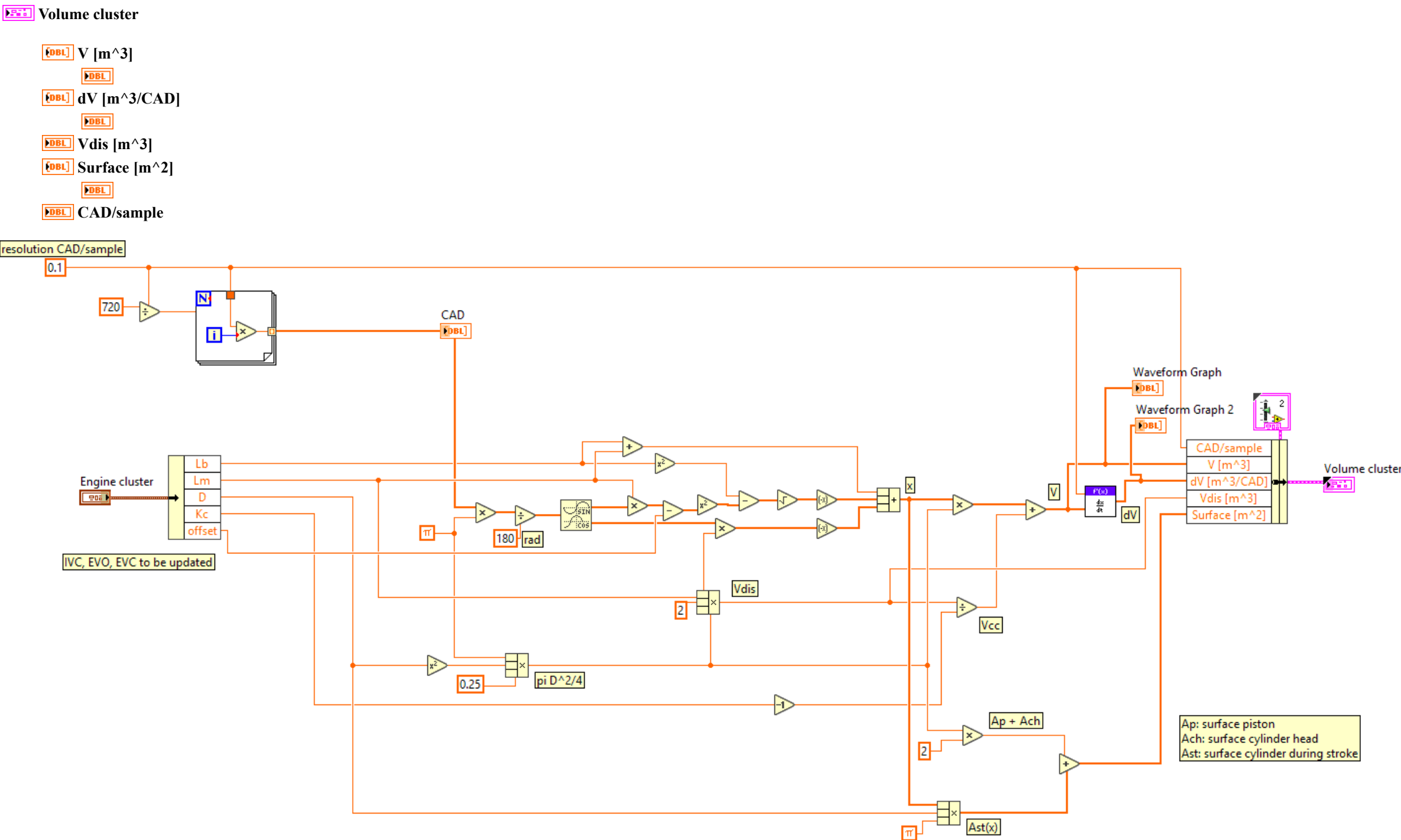
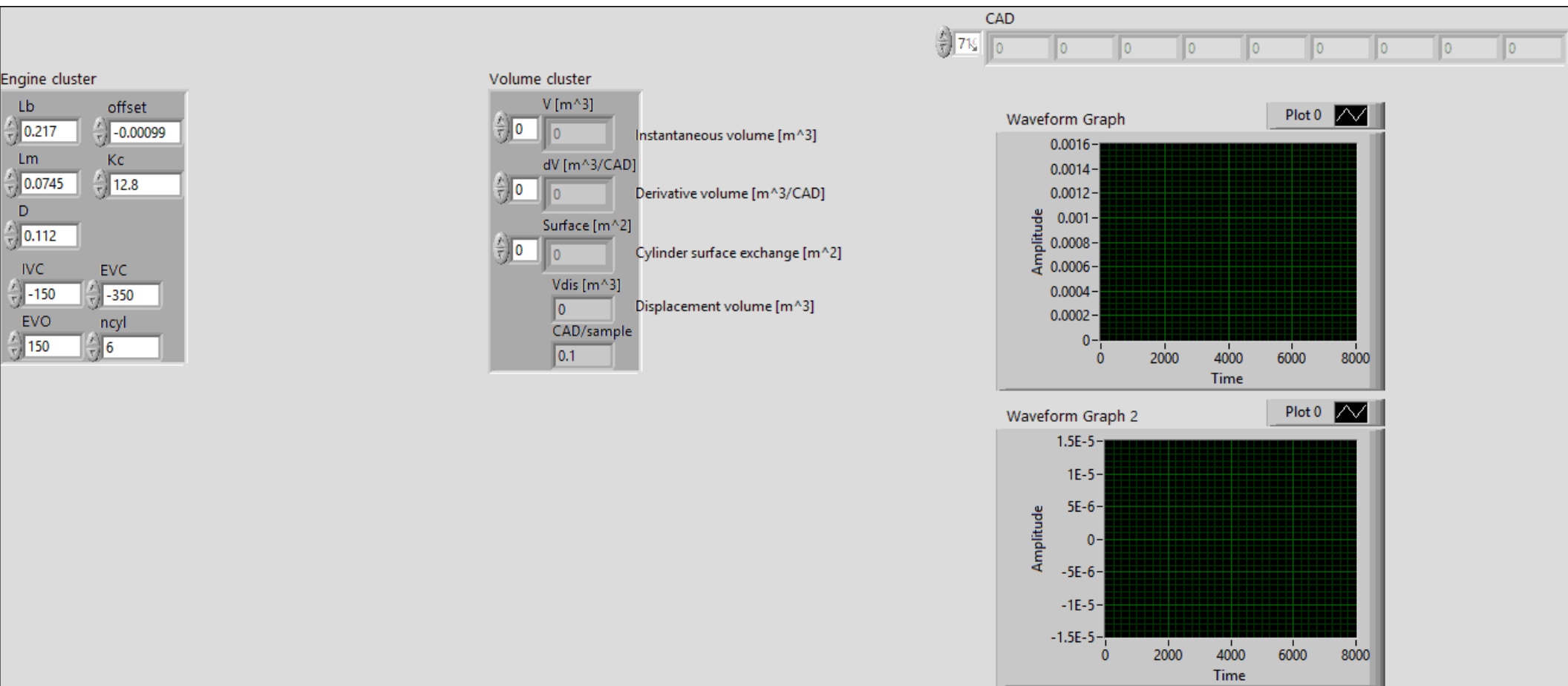
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\3filter.lib\Zero Phase Filter (Cascade, DBL).vi

NI_AALBase.lib:Zero Phase Filter.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\3filter.lib\Zero Phase Filter.vi

APC_9049_HRL_MAPO.vi

Pcyl [4] [1] MAPO
IIR Filter Cluster [3] [0] PcylHPfilt

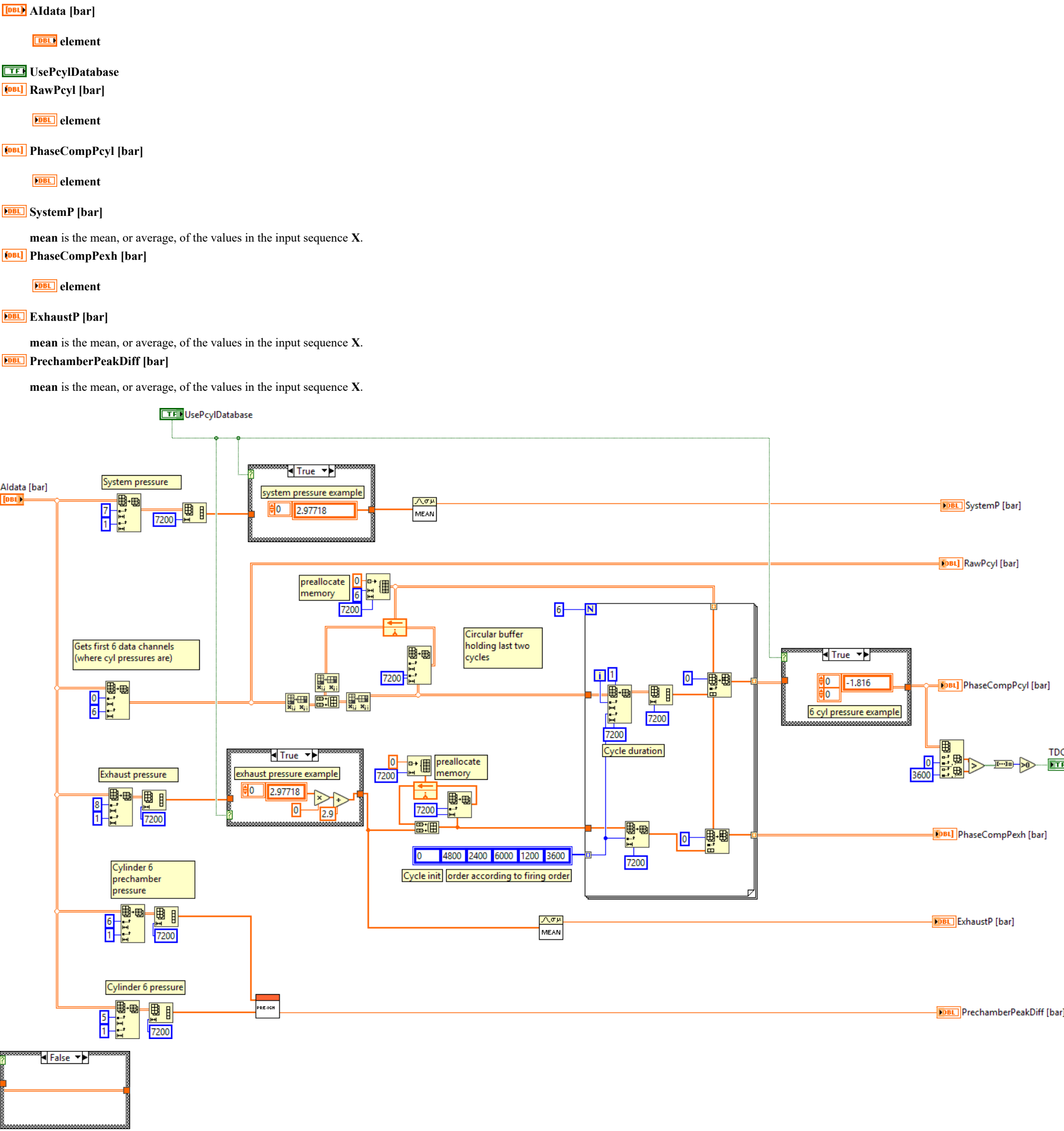
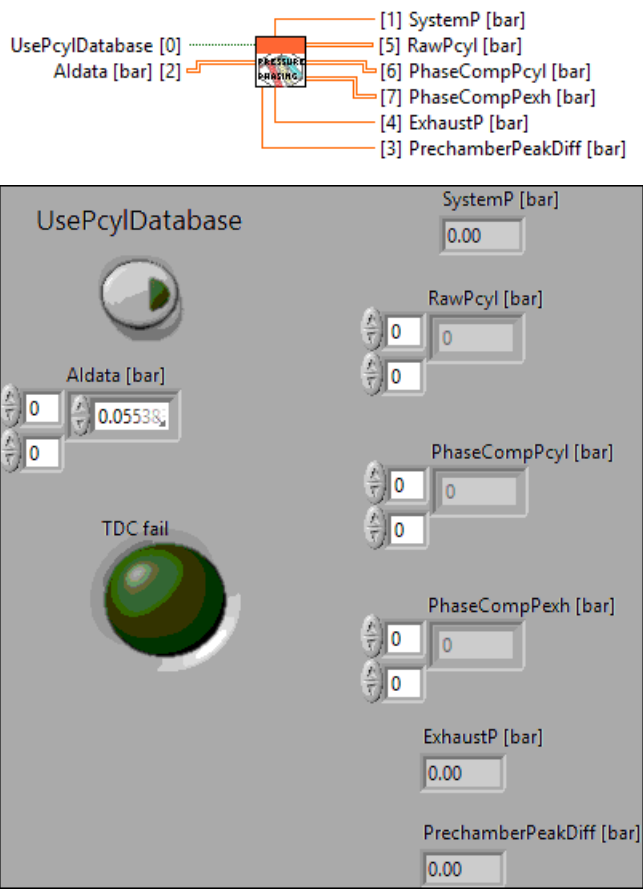


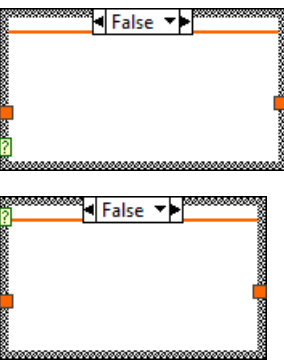
APC_volume.vi
C:\LabVIEW PROJECT\MONARCH\controls\APC_volume.vi
APC_CAD_derivative.vi
C:\LabVIEW PROJECT\MONARCH\support\APC_CAD_derivative.vi

APC_9049_PPhaseCorrection.vi

This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

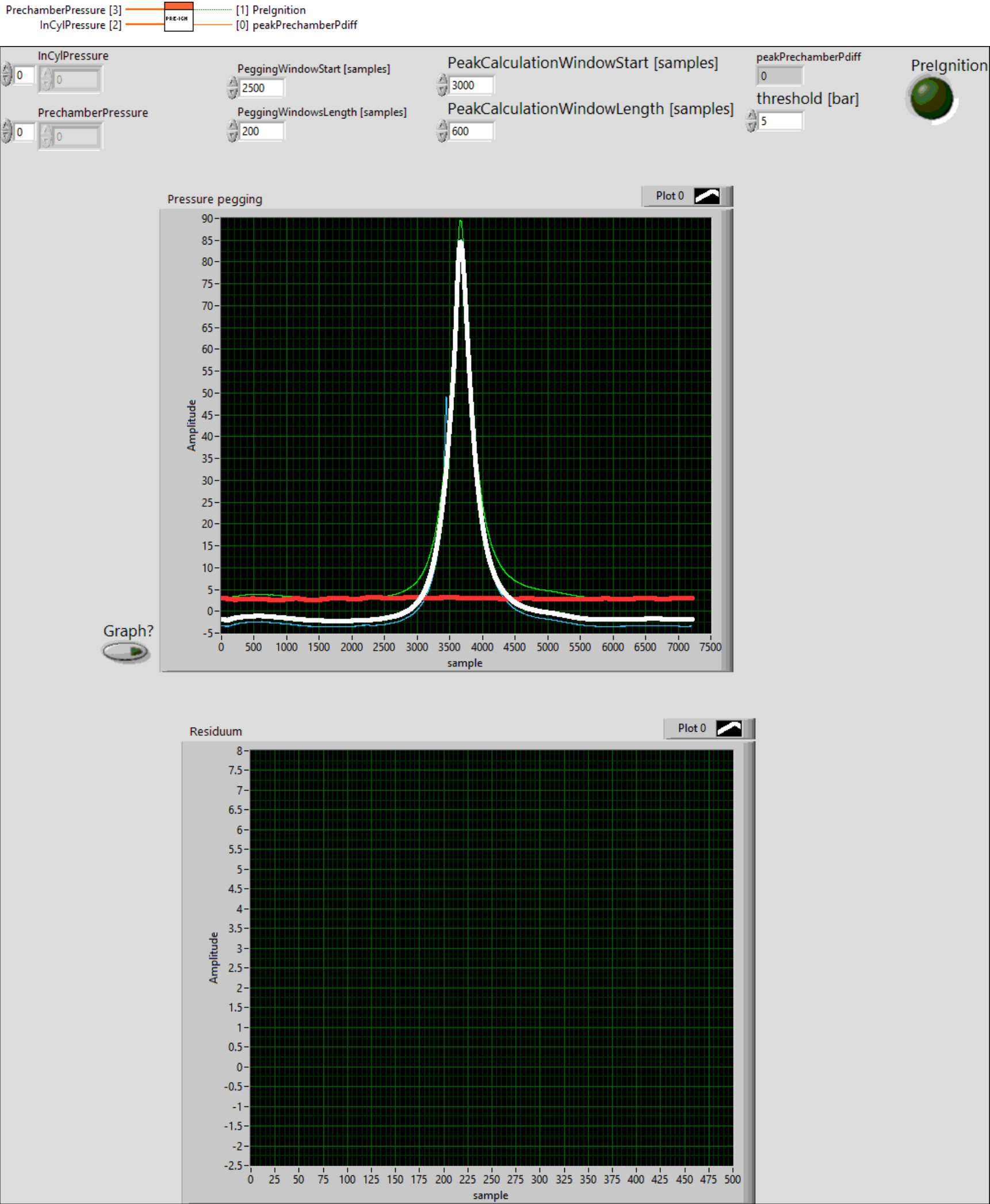
The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.



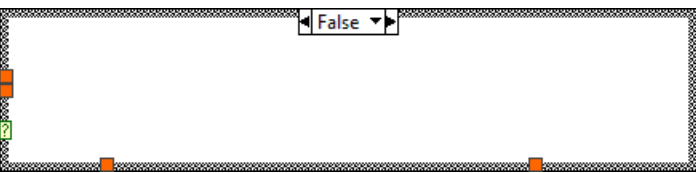
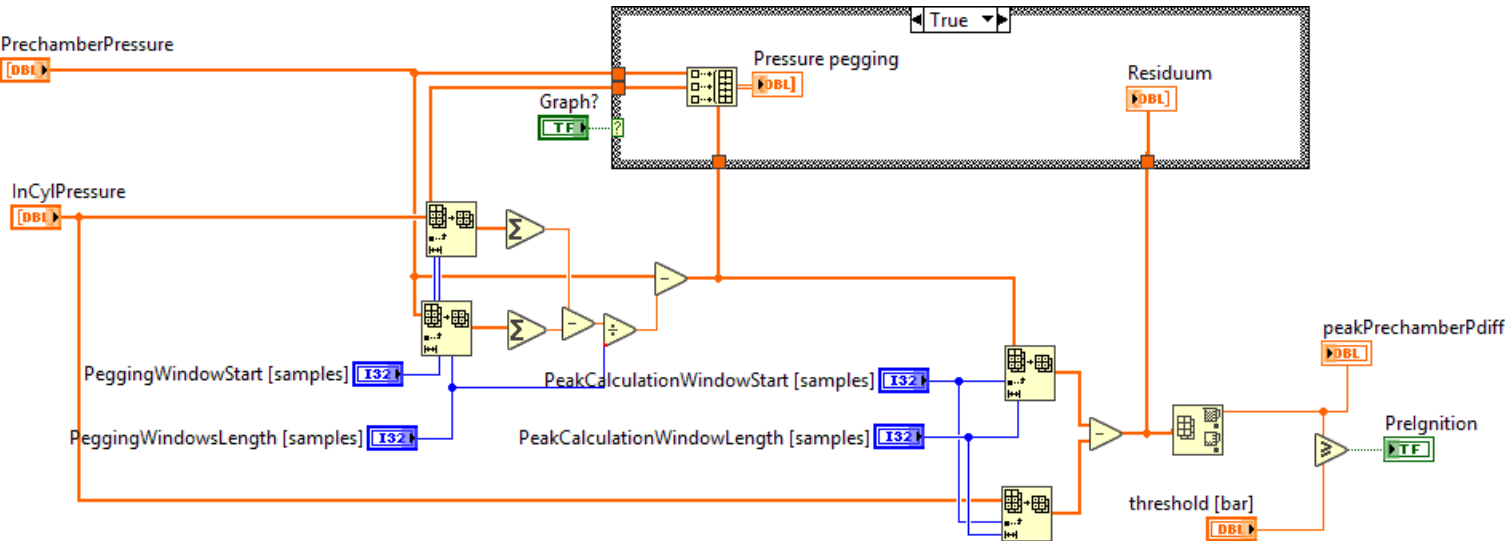


NI_AALBase.lvlib:Mean (DBL).vi
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\bascanly.llb\Mean (DBL).vi
NI_AALBase.lvlib:Mean.vi
C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\bascanly.llb\Mean.vi
APC_9049_HRL_preChamber.vi
C:\LabVIEW PROJECT\MONARCH\APC_9049_HRL_preChamber.vi

APC_9049_HRL_preChamber.vi



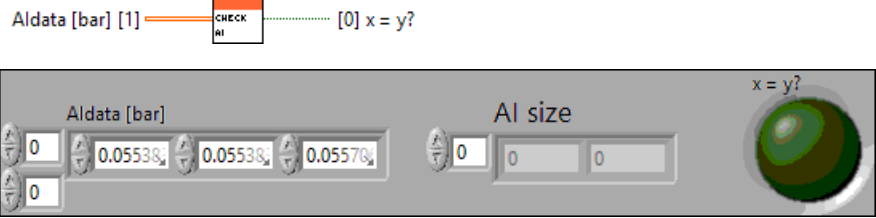
InCylPressure
Numeric
PrechamberPressure
Prelgnition
peakPrechamberPdiff



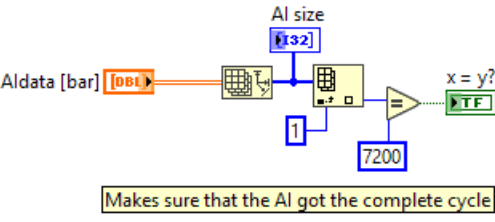
APC_9049_checkAI.vi

This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.



Aldata [bar]
element
x = y?



APC_CylDagnosticsCluster.ctl



CylinderDiagnosticsCluster

samples for running IMEP std

100

MaxDevFromExpectedIMEPError [bar]

2

MaxDevFromExpectedIMEPWarning [bar]

2

MaxDevFromAvgError [bar]

2

MaxDevFromSelfAvgWarning [bar]

2

MaxDevFromSelfAvgError [bar]

2

MaxDevFromAvgWarning [bar]

2

MaxIMEPstdWarning [bar]

0.5

MaxIMEPstdError [bar]

0.5

MAPOmaxError [bar/CAD]

1

MAPOmaxWarning [bar/CAD]

1

CA50maxError [CADATDC]

15

CA50maxWarning [CADATDC]

15

MaxPCylIMaxError [bar]

120

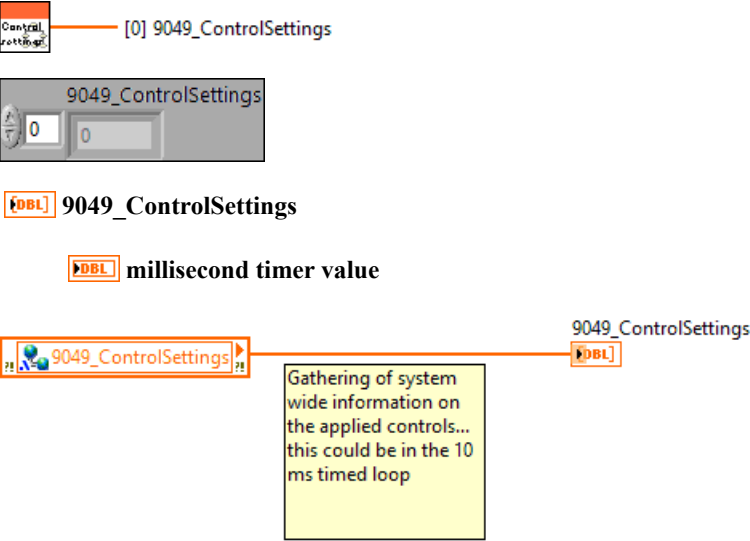
MaxPCylIMaxWarning [bar]

120

APC_9049_ControlSettingsPolling.vi

This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

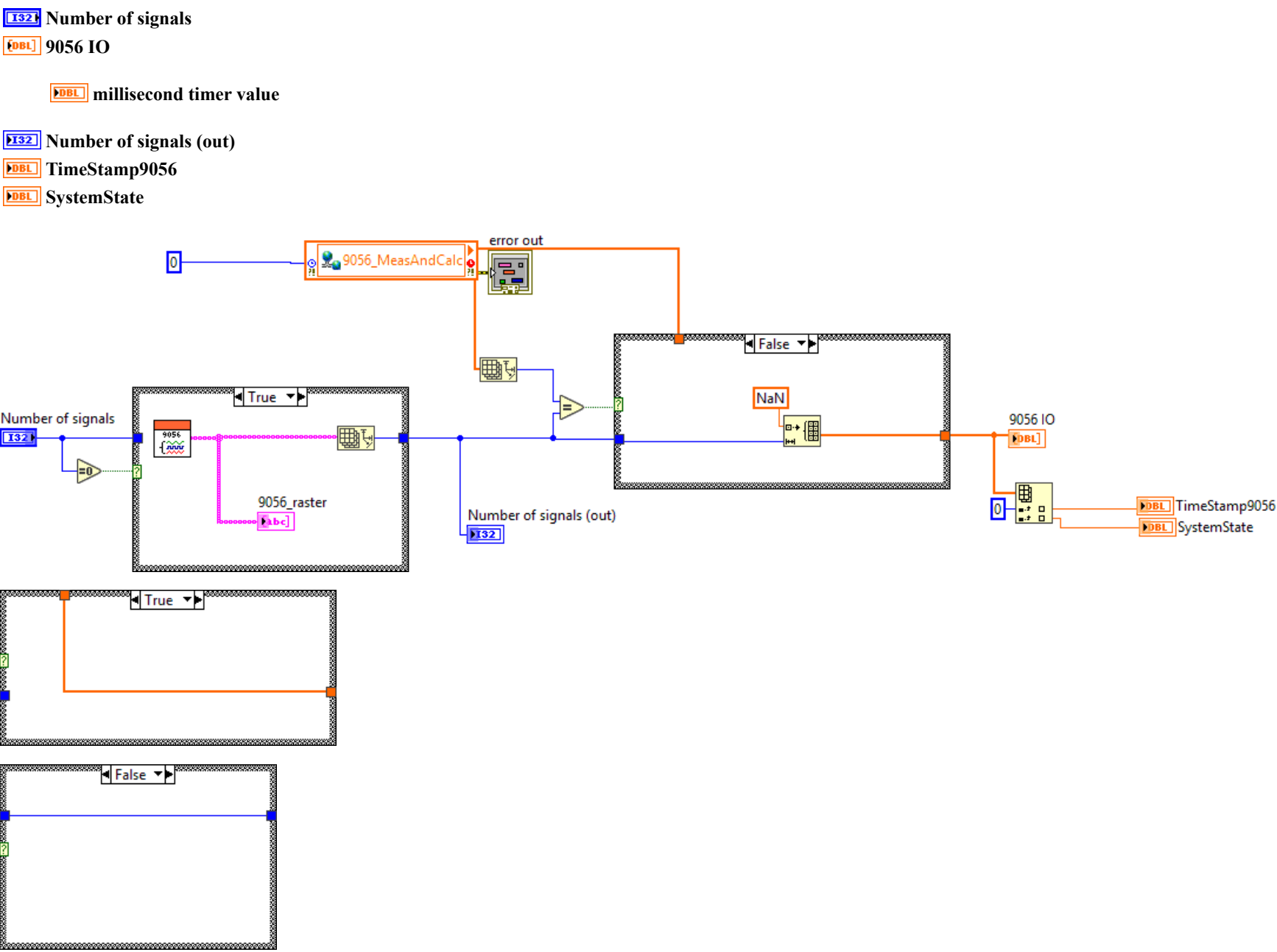
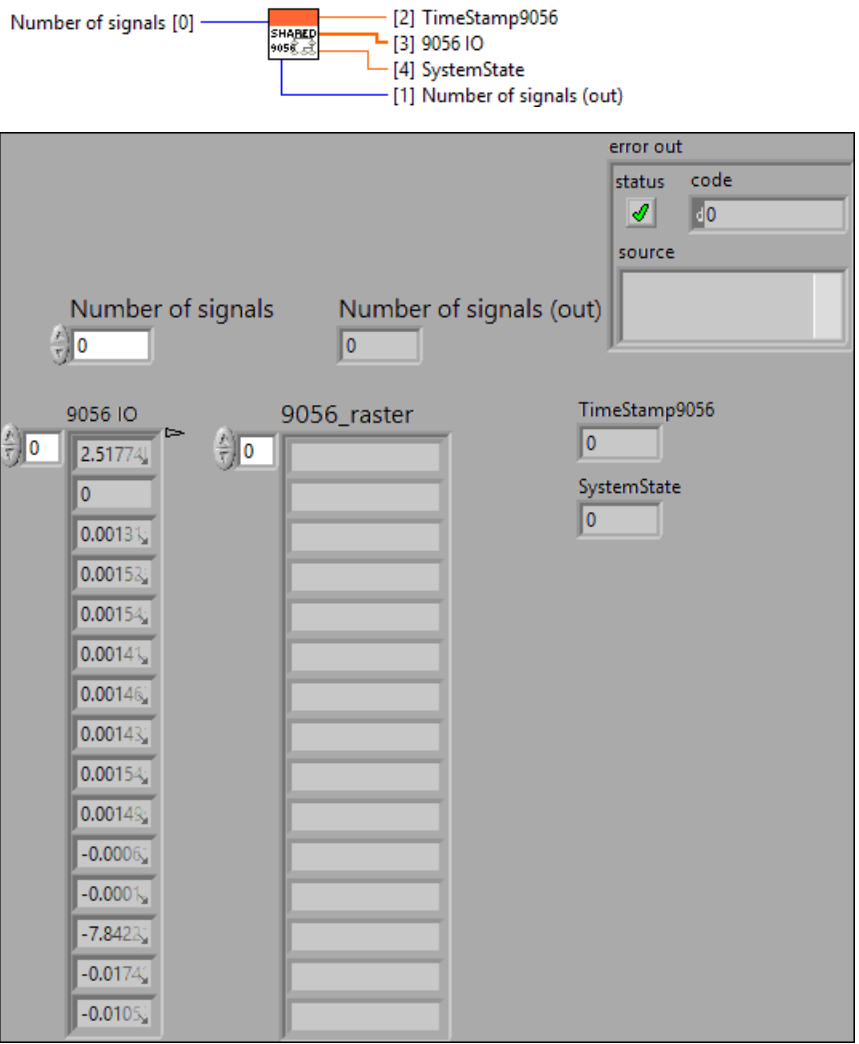
The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.



APC_9049_9056SharedVarPolling.vi

This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

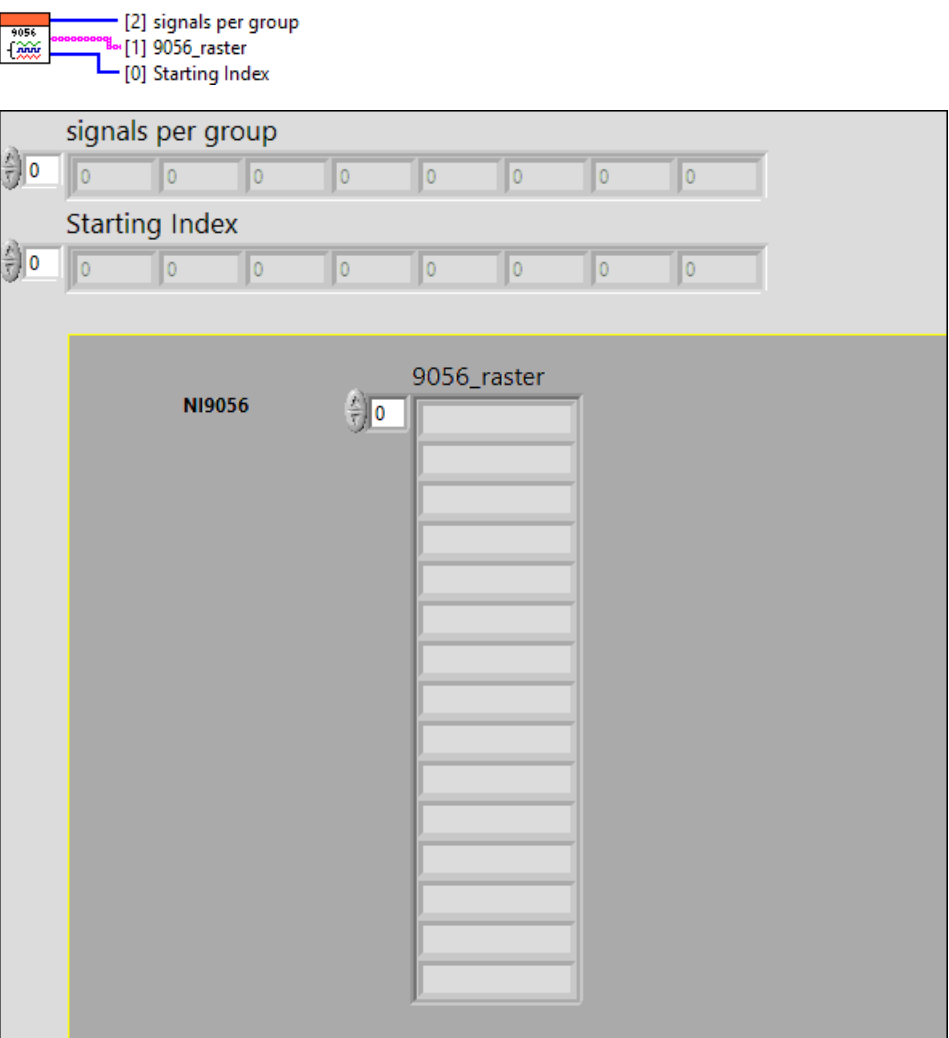
The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.



APC_9056_Signals.vi

C:\LabVIEW PROJECT\MONARCH\APC_9056_Signals.vi

APC_9056_Signals.vi



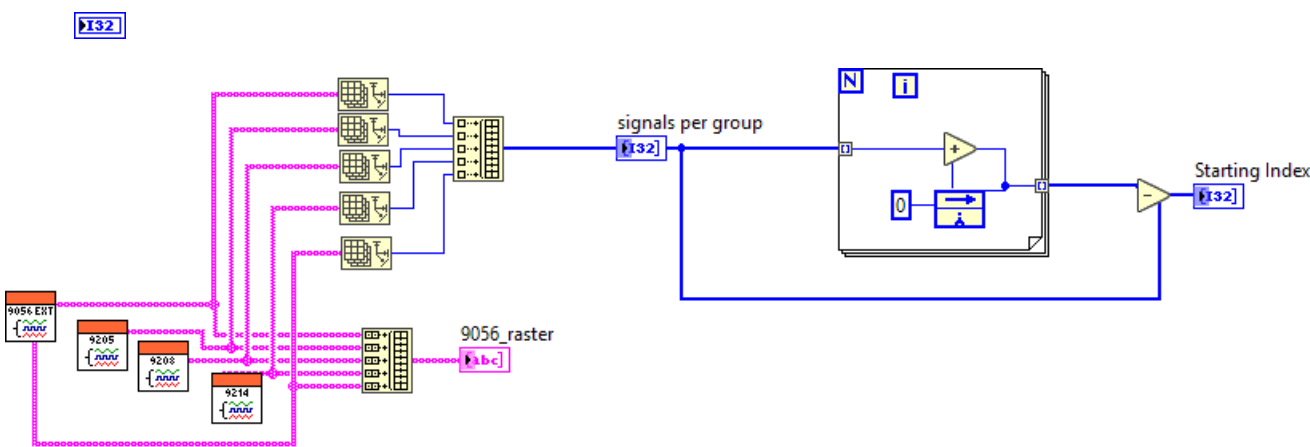
9056_raster

name to assign

signals per group

Starting Index

[I32] Starting Index



APC_9056_NI9214conf.vi

C:\LabVIEW PROJECT\MONARCH\APC_9056_NI9214conf.vi

APC_9056_NI9208conf.vi

C:\LabVIEW PROJECT\MONARCH\APC_9056_NI9208conf.vi

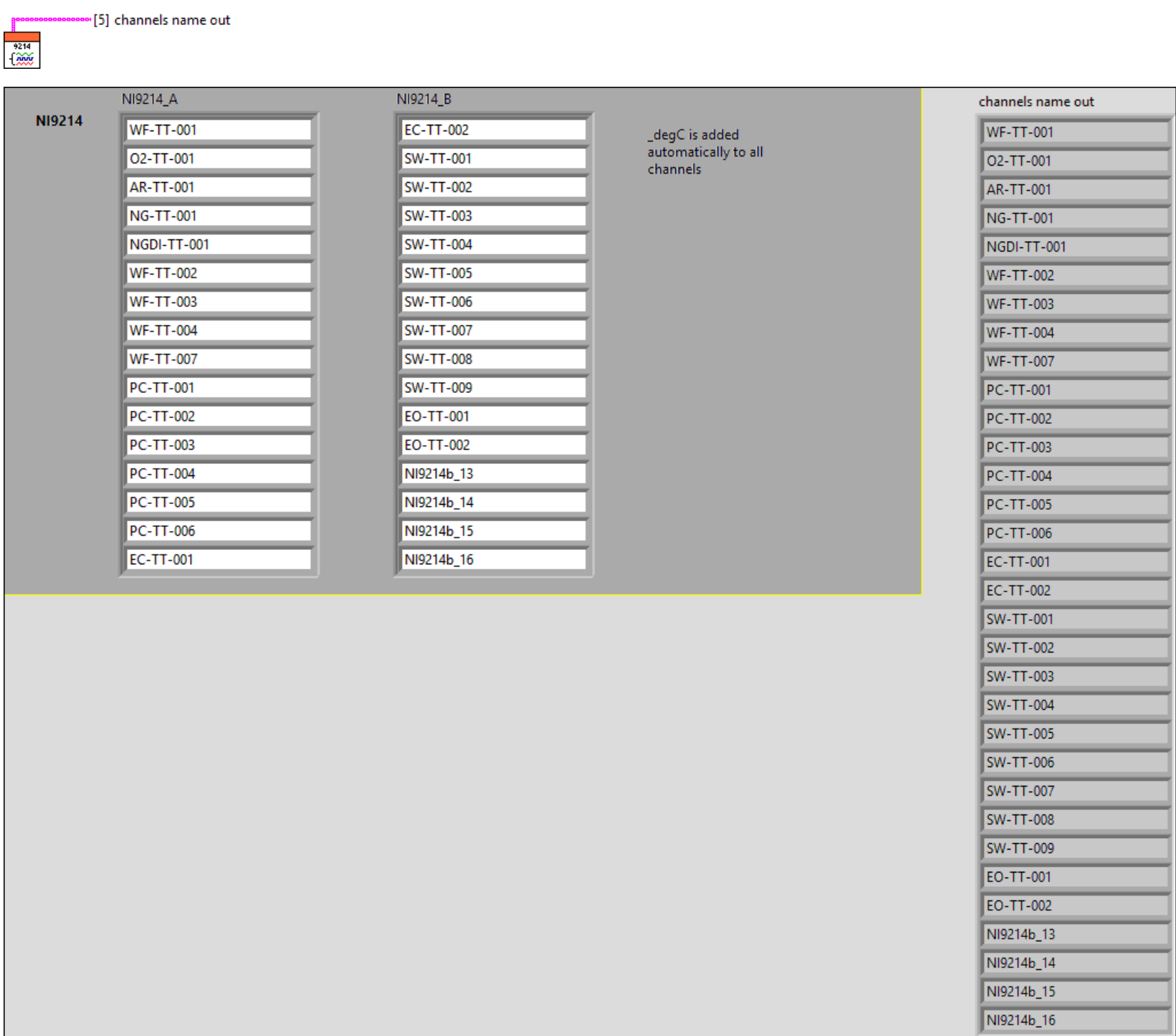
APC_9056_NI9205conf.vi

C:\LabVIEW PROJECT\MONARCH\APC_9056_NI9205conf.vi

APC_9056_additionalSignals.vi

C:\LabVIEW PROJECT\MONARCH\APC_9056_additonalSignals.vi

APC_9056_NI9214conf.vi

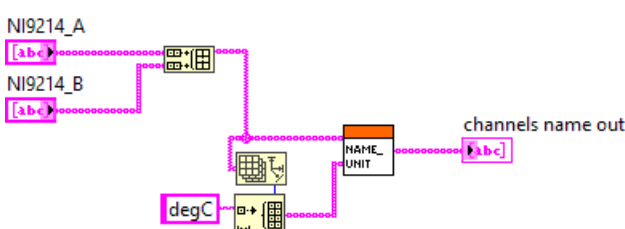


{abc} channels name out

name to assign specifies a name to assign to the virtual channel this VI creates. If you do not wire a value to this input, NI-DAQmx uses the physical channel name as the virtual channel name.

 name to assign

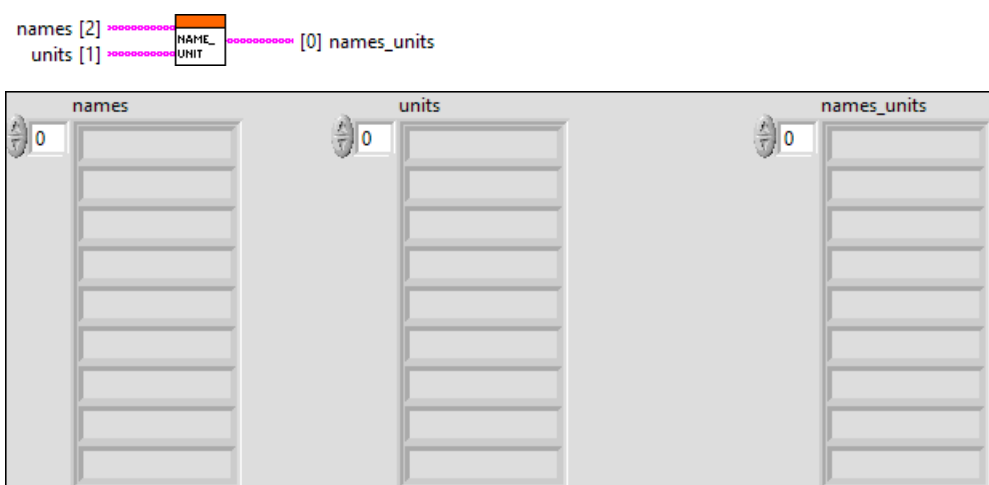
name to assign specifies a name to assign to the virtual channel this VI creates. If you do not wire a value to this input, NI-DAQmx uses the physical channel name as the virtual channel name.



APC_AppendUnits.vi

C:\LabVIEW PROJECT\MONARCH\APC_AppendUnits.vi

APC_AppendUnits.vi



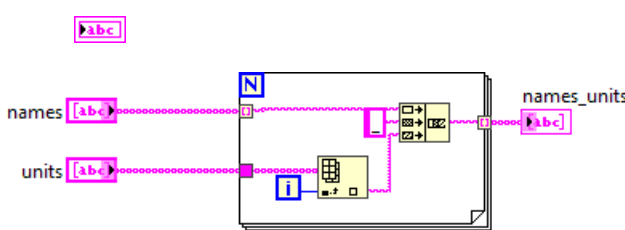
[abc] names



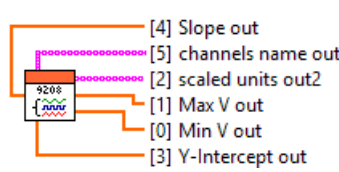
[abc] units

abc scaled units

[abc] names_units



APC 9056 NI9208conf.vi



N19208	channels name		scaled units	Slope	Y-intercept	Max I	Min I	channels name out	scaled units	Sample out	Y-Intercept out	Max V out	Min V out
	NG-FT-001		kgs	1.00	0.00	0.02	-0.02	NG-FT-001_kgs	kgs	1.00	0.00	0.02	-0.02
	NG-PT-001		bar	1.00	0.00	0.02	-0.02	NG-PT-001_bar	bar	1.00	0.00	0.02	-0.02
	NGDI-PT-001		bar	1.00	0.00	0.02	-0.02	NGDI-PT-001_bar	bar	1.00	0.00	0.02	-0.02
	O2-FC-001		kgs	1.00	0.00	0.02	-0.02	O2-FC-001_kgs	kgs	1.00	0.00	0.02	-0.02
	NG-FC-001		kgs	1.00	0.00	0.02	-0.02	NG-FC-001_kgs	kgs	1.00	0.00	0.02	-0.02
	NGDI-FT-001		kgs	1.00	0.00	0.02	-0.02	NGDI-FT-001_kgs	kgs	1.00	0.00	0.02	-0.02
	WF-PT-004		bar	1.00	0.00	0.02	-0.02	WF-PT-004_bar	bar	1.00	0.00	0.02	-0.02
	O2-PT-001		bar	1.00	0.00	0.02	-0.02	O2-PT-001_bar	bar	1.00	0.00	0.02	-0.02
	AR-PT-001		bar	1.00	0.00	0.02	-0.02	AR-PT-001_bar	bar	1.00	0.00	0.02	-0.02
	AR-FT-001		kgs	1.00	0.00	0.02	-0.02	AR-FT-001_kgs	kgs	1.00	0.00	0.02	-0.02
	WF-PT-016		bar	1.00	0.00	0.02	-0.02	WF-PT-016_bar	bar	1.00	0.00	0.02	-0.02
	WF-PT-014		bar	1.00	0.00	0.02	-0.02	WF-PT-014_bar	bar	1.00	0.00	0.02	-0.02
	SW-FT-001		kgs	1.00	0.00	0.02	-0.02	SW-FT-001_kgs	kgs	1.00	0.00	0.02	-0.02
	SW-FT-002		kgs	1.00	0.00	0.02	-0.02	SW-FT-002_kgs	kgs	1.00	0.00	0.02	-0.02
	SW-FT-003		kgs	1.00	0.00	0.02	-0.02	SW-FT-003_kgs	kgs	1.00	0.00	0.02	-0.02
	SW-FT-004		kgs	1.00	0.00	0.02	-0.02	SW-FT-004_kgs	kgs	1.00	0.00	0.02	-0.02

{abc} channels name out

name to assign specifies a name to assign to the virtual channel this VI creates. If you do not wire a value to this input, NI-DAQmx uses the physical channel name as the virtual channel name.

 name to assign

name to a

[DBL] Slope out

► OBL **Numeric**

[DBL] Y-Intercept out

DL Numeric

[DBL] Max V out

 AI.Max

(DBL) Min V out

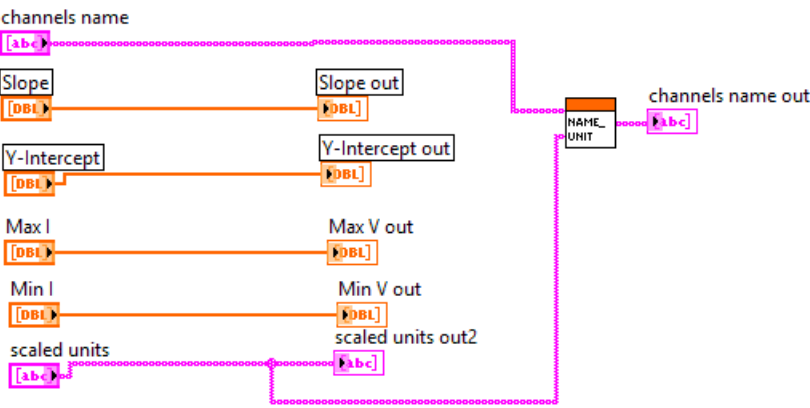
 AI.Max

{abc} scaled units out

scaled units is the units to use for the scaled value. You can use an arbitrary string. LabVIEW uses the units to label a graph or chart.

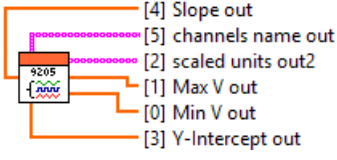
 scaled units

scaled units is the units to use for the scaled value. You can use an arbitrary string. LabVIEW uses the units to label a graph or chart.



APC_AppendUnits.vi
C:\LabVIEW PROJECT\MONARCH\APC_AppendUnits.vi

APC_9056_NI9205conf.vi



NI9205	channels name	scaled units	Slope	Y-Intercept	Max V	Min V	channels name out	scaled units	Slope out	Y-Intercept out	Max V out	Min V out
	WF-PT-001	bar	1.00	0.00	10	-10	WF-PT-001_bar	bar	1.00	0.00	10	-10
	WF-OA-001-O2	perc	1.00	0.00	10	-10	WF-OA-001-O2_perc	perc	1.00	0.00	10	-10
	WF-OA-001-O2corr	perc	1.00	0.00	10	-10	WF-OA-001-O2corr_perc	perc	1.00	0.00	10	-10
	WF-OA-001-Lambda	r	1.00	0.00	10	-10	WF-OA-001-Lambda_r	r	1.00	0.00	10	-10
	WF-OA-002-O2	perc	1.00	0.00	10	-10	WF-OA-002-O2_perc	perc	1.00	0.00	10	-10
	WF-OA-002-Lambda	r	1.00	0.00	10	-10	WF-OA-002-Lambda_r	r	1.00	0.00	10	-10
	WF-OA-002-NOx	ppm	1.00	0.00	10	-10	WF-OA-002-NOx_ppm	ppm	1.00	0.00	10	-10
	EO-PT-001	bar	1.00	0.00	10	-10	EO-PT-001_bar	bar	1.00	0.00	10	-10
	TORQUE	Nm	1.00	0.00	10	-10	TORQUE_Nm	Nm	1.00	0.00	10	-10
	NI9205_9	V	1.00	0.00	10	-10	NI9205_9_V	V	1.00	0.00	10	-10
	NI9205_10	V	1.00	0.00	10	-10	NI9205_10_V	V	1.00	0.00	10	-10
	NI9205_11	V	1.00	0.00	10	-10	NI9205_11_V	V	1.00	0.00	10	-10
	NI9205_12	V	1.00	0.00	10	-10	NI9205_12_V	V	1.00	0.00	10	-10
	NI9205_13	V	1.00	0.00	10	-10	NI9205_13_V	V	1.00	0.00	10	-10
	NI9205_14	V	1.00	0.00	10	-10	NI9205_14_V	V	1.00	0.00	10	-10
	NI9205_15	V	1.00	0.00	10	-10	NI9205_15_V	V	1.00	0.00	10	-10

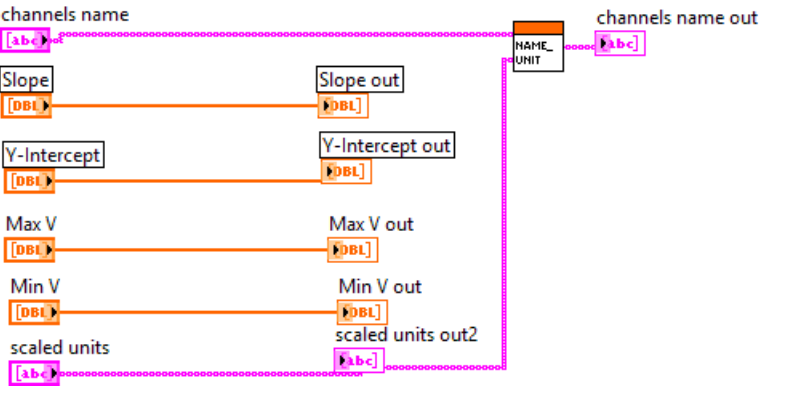
channels name out
name to assign specifies a name to assign to the virtual channel this VI creates. If you do not wire a value to this input, NI-DAQmx uses the physical channel name as the virtual channel name.

name to assign
name to assign specifies a name to assign to the virtual channel this VI creates. If you do not wire a value to this input, NI-DAQmx uses the physical channel name as the virtual channel name.

Slope out
Numeric
Y-Intercept out
Numeric
Max V out
AI.Max
Min V out
AI.Max

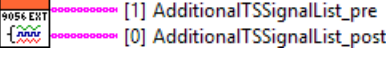
scaled units out2
scaled units is the units to use for the scaled value. You can use an arbitrary string. LabVIEW uses the units to label a graph or chart.
scaled units

scaled units is the units to use for the scaled value. You can use an arbitrary string. LabVIEW uses the units to label a graph or chart.



APC_AppendUnits.vi
C:\LabVIEW PROJECT\MONARCH\APC_AppendUnits.vi

APC_9056_additonalSignals.vi



NAMING: for calculated and OUTPUT variables , include units on variable names.						
	_bar					
	_degC	(°C)				
	_perc	(%)				
	_rpm					
	_r	(ratio)				
	_b	(bool)				
	_u64					
NI9056	Signal raster previous to the AI	NI9266	NI9375_DO	NI9375_DI	NI9264	From PC
	TimeStamp9056_ms	O2-FC-001-REF_mA	NG-FV-001_b	NI9375_AI0_b	NI9264_AO0_V	AIC103_CO2_ConcTarget_ack
	SystemState_i8	AR-PC-002-REF_mA	O2-PV-004_b	NI9375_AI1_b	NI9264_AO1_V	AT103_CO2Conc
		NG-FC-001-REF_mA	AR-PV-004_b	NI9264_AO2_b	NI9375_AI2_b	PV103_PressTarget_kPa_ack
		EC-FC-001_mA	WF-FC-001_b	NI9375_AI3_b	NI9264_AO3_V	PT103_Pressure_kPa
		SW-FC-004-REF_mA	WF-FC-002_b	NI9264_AO4_b	NI9375_AI4_b	PV103_VlvPosTarget_pc_ack
		SW-FC-009_mA	WF-FC-003_b	NI9375_AI5_b	NI9264_AO5_V	PV103_VlvPos_pc_fb
		DYNO-REF_mA	NI9375_AO6_b	NI9375_AI6_b	NI9264_AO6_V	AIC201_CO2_ConcTarget_ack
		NI9266_AO7_mA	NI9375_AO7_b	NI9375_AI7_b	NI9264_AO7_V	AT201_CO2Conc
			NI9375_AO8_b	NI9375_AI8_b	NI9264_AO8_V	FV301_FlowTarget_kgh_ack
			NI9375_AO9_b	NI9264_AO9_V	NI9375_AI9_b	FT301_Flow_kgh
			NI9375_AO10_b	NI9375_AI10_b	NI9264_AO10_V	FV301_VlvPosTarget_pc_ack
			NI9375_AO11_b	NI9375_AI11_b	NI9264_AO11_V	FV301_VlvPos_pc_fb
			NI9375_AO12_b	NI9375_AI12_b	NI9264_AO12_V	PV300_PressTarget_kPa_ack
			NI9375_AO13_b	NI9375_AI13_b	NI9264_AO13_V	PT300_Pressure_kPa
			NI9375_AO14_b	NI9375_AI14_b	NI9264_AO14_V	PV300_VlvPosTarget_pc_ack
			NI9375_AO15_b	NI9375_AI15_b	NI9264_AO15_V	PV300_VlvPos_pc_fb
						C101_CompSpeed_rpm
						PT104_Pressure_kPa
						PT106_Pressure_kPa
						PLC_Manual_Control
						MTR_State
						MTR_HeartBeat
						PV103_Ctrl_Mode_act
						FV301_Ctrl_Mode_act
						PV300_Ctrl_Mode_act
						MTR_Diagnostic_Status

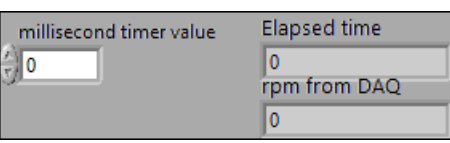
AdditionalTSSignalList_pre
AdditionalTSSignalList_post
Signal raster previous to the AI
AdditionalTSSignalList_pre
AdditionalTSSignalList_post

APC_9049_rpm.vi

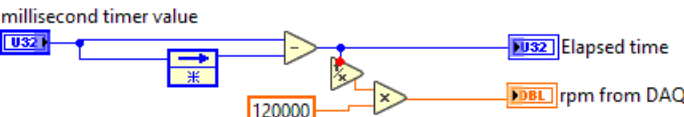
This example demonstrates the procedure required to apply different NI-DAQmx Custom Scales to each virtual channel in a task. A DAQmx Task Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition >> DAQmx Advanced Task Options) is used to obtain the task's virtual channels. A DAQmx Channel Property Node (Functions >> NI Measurements >> DAQmx - Data Acquisition) is then used to set the custom scale for each virtual channel in the task.

The "DAQmx Create Scale" VI creates a custom scale for each channel in the channel list. *Note, make sure your Minimum and Maximum channel parameters are consistent with the range the scale will create.

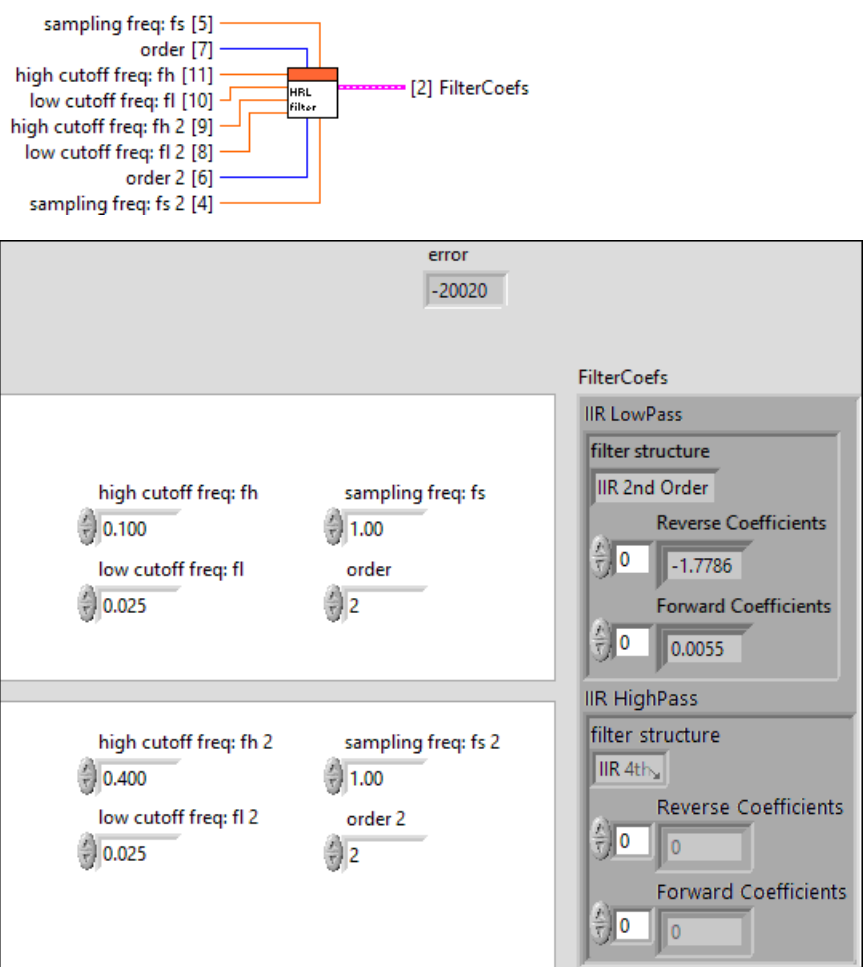
millisecond timer value [1] rpm from DAQ



millisecond timer value
rpm from DAQ



APC_9049_HRL_CreateFilter.vi



order

order is the order of the IIR filter and must be greater than zero. If **order** is less than or equal to zero, the VI sets **Reverse Coefficients** and **Forward Coefficients** to empty arrays and returns an error.

low cutoff freq: fl

low cutoff freq: fl is the low cutoff frequency in Hz and must observe the Nyquist criterion.

high cutoff freq: fh

high cutoff freq: fh is the high cutoff frequency in Hz. The default is 0.45 Hz.

sampling freq: fs

sampling freq: fs is the sampling frequency in Hz and must be greater than zero. The default is 1.0 Hz.

order 2

order is the order of the IIR filter and must be greater than zero. If **order** is less than or equal to zero, the VI sets **Reverse Coefficients** and **Forward Coefficients** to empty arrays and returns an error.

low cutoff freq: fl 2

low cutoff freq: fl is the low cutoff frequency in Hz and must observe the Nyquist criterion.

high cutoff freq: fh 2

high cutoff freq: fh is the high cutoff frequency in Hz. The default is 0.45 Hz.

sampling freq: fs 2

sampling freq: fs is the sampling frequency in Hz and must be greater than zero. The default is 1.0 Hz.

FilterCoefs

IIR LowPass

IIR Filter Cluster contains the cascaded form of IIR filter coefficients.

filter structure

filter structure indicates either IIR second-order or IIR fourth-order filter stages.

Reverse Coefficients

Reverse Coefficients is the reverse coefficients of the IIR cascade filter.

Reverse Coefficients is the reverse coefficients of the IIR cascade filter.

Forward Coefficients

Forward Coefficients is the forward coefficients of the IIR cascade filter.

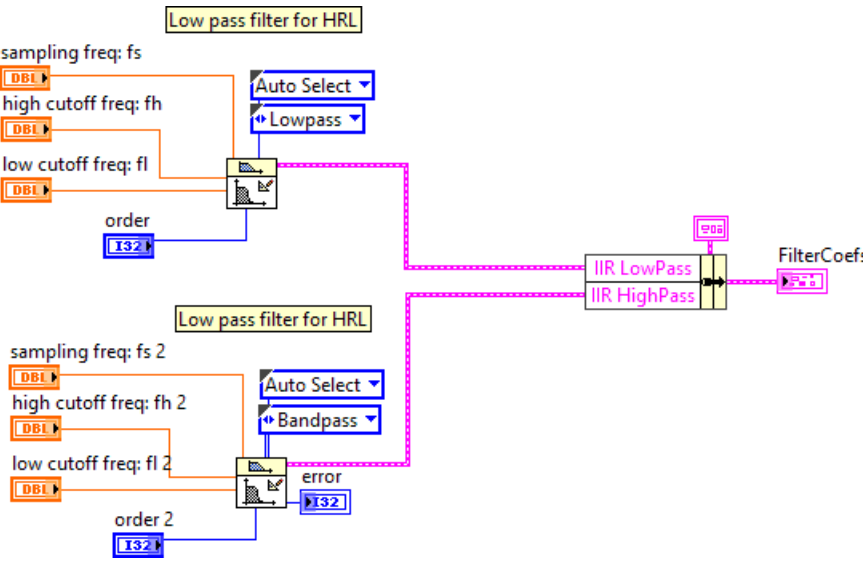
Forward Coefficients is the forward coefficients of the IIR cascade filter.

IIR HighPass

filter structure

Reverse Coefficients

Forward Coefficients



NT_AALBase.lvlib:Butterworth Coefficients.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\3filter.lib\Butterworth Coefficients.vi

NT_AALBase.lvlib:Filter Structure Type.ctl

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\3filter.lib\Filter Structure Type.ctl

NT_AALBase.lvlib:Filter Passband Type.ctl

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Analysis\3filter.lib\Filter Passband Type.ctl

APC_DAOqueues.ctl

